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Analysis and Optimization of Ammonia-Hydrogen Reaction Mechanisms

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1. Introduction ConclusionReferencesIntroduction

In recent years, research into carbon-free fuels such as hydrogen and ammonia has intensified in pursuit of a decarbonized society. Ammonia, in particular, is attracting attention as an excellent hydrogen energy carrier because it has a high hydrogen storage density and, with a boiling point and condensation temperature similar to those of propane, is easy to store and transport. At the same time, research is also underway to use ammonia directly as a fuel, as it does not produce CO_2 during combustion. However, compared to methane, ammonia has a significantly lower laminar combustion velocity, raising concerns about reduced combustion efficiency and flame extinction due to insufficient residence time within the combustor [1-2]. To improve this low combustibility, co-combustion with high-flame-velocity fuels such as hydrocarbons (e.g., methane) or hydrogen has been proposed [3-4], and it has been reported that this approach yields higher laminar combustion velocities than those of a pure ammonia flame [5].

However, in $\text{NH}_3/\text{CH}_4/\text{Air}$ and $\text{NH}_3/\text{H}_2/\text{Air}$ flames, high flame temperatures and the abundant generation of H and OH radicals compared to pure NH_3/Air flames result in high NO_x formation, which poses a challenge [6-8]. Understanding the combustion mechanism is essential for the development of NO_x suppression technologies, and to date, the development and optimization of reaction mechanisms have been focused primarily on $\text{NH}_3/\text{CH}_4/\text{Air}$ and $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

For example, Okafor et al. [9-10] developed a detailed reaction mechanism (Okafor Detailed Mech [9]) and a simplified reaction mechanism (Okafor Reduced Mech [10]) with high reproducibility in $\text{NH}_3/\text{CH}_4/\text{Air}$ flames. According to Okafor et al.'s analysis, no significant direct interaction was observed between the oxidation pathways of ammonia and methane during co-combustion; instead, the two are said to be limited to an "indirect coupling" in which they primarily share O, H, and OH radical pools [9].

On the other hand, it has been reported that in $\text{NH}_3/\text{H}_2/\text{Air}$ flames, the oxidation pathways of both NH_3 and H_2 are directly linked, resulting in extremely complex interactions. Gotama et al. have conducted a detailed analysis of the relationship between the amount of hydrogen added and the reaction kinetics in $\text{NH}_3/\text{H}_2/\text{Air}$ flames. According to their report, it is suggested that different kinetic interactions exist between hydrogen and each NH_3 -derived species depending on the concentration conditions [11].

However, as mentioned above, many existing reaction mechanisms are currently optimized individually for specific combustion conditions, such as $\text{NH}_3/\text{H}_2/\text{Air}$ flames, $\text{NH}_3/\text{CH}_4/\text{Air}$ flames, or pure NH_3/Air flames; no reaction mechanism has been reported that is simultaneously optimized for both $\text{NH}_3/\text{H}_2/\text{Air}$ and $\text{NH}_3/\text{CH}_4/\text{Air}$ flames [12]. For example, while the reaction mechanism developed by Okafor et al. can accurately predict the laminar combustion rate of $\text{NH}_3/\text{CH}_4/\text{Air}$ flames, it has been reported to underestimate the laminar combustion rate in $\text{NH}_3/\text{H}_2/\text{Air}$ flames [13]. Flexible combustion prediction based on fuel composition is required for the co-combustion of ammonia, but a unified reaction model capable of accurately reproducing both hydrocarbon and hydrogen co-combustion has not yet been established. Therefore, in this study, based on the Okafor Reduced Mech [10], which exhibits high accuracy for $\text{NH}_3/\text{CH}_4/\text{Air}$ flames, we construct a reaction mechanism capable of accurately predicting the laminar combustion rate under both $\text{NH}_3/\text{CH}_4/\text{Air}$ and $\text{NH}_3/\text{H}_2/\text{Air}$ flame conditions, with the aim of identifying the key combustion mechanisms, particularly in $\text{NH}_3/\text{H}_2/\text{Air}$ flames

2. Numerical Simulation

In this study, simulations of one-dimensional adiabatic premixed laminar flames were performed using ANSYS Chemkin-PRO 2019 R3 [14] and ANSYS Chemkin 2025 R2 [15]. Hereinafter, ANSYS Chemkin-PRO 2019 R3 [14] and ANSYS Chemkin 2025 R2 [15] will be referred to as “Chemkin.” First, to evaluate the predictive accuracy of the reaction mechanisms in an $\text{NH}_3/\text{H}_2/\text{Air}$ flame, we compared the calculated values of the laminar combustion rate under stoichiometric conditions for an $\text{NH}_3/\text{H}_2/\text{Air}$ flame (excluding flame stretching) with experimental values reported in other literature. Furthermore, to analyze the characteristics of each reaction mechanism in detail, sensitivity analyses were performed on the rate of production (ROP) and the sensitivity of each reaction to the laminar combustion rate. The reaction mechanisms used were the **the**, Nakamura Mech [16], Otomo Mech [17], Okafor Reduced Mech [10], and Gotama Detailed Mech [11].

Reaction pathway analysis using the rate of production (ROP) of chemical species identified the cause of the underestimation of the laminar combustion rate of the $\text{NH}_3/\text{H}_2/\text{Air}$ flame by the **Okafor Reduced Mech**. Next, to enable accurate prediction of the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames under stoichiometric conditions, we constructed a new reaction mechanism by refining the reaction rate constants, thermodynamic data, and transport properties of the **Okafor Reduced Mech** [10]. Hereinafter, the reaction mechanism proposed in this study will be referred to as “**present mech.**”

To verify the predictive accuracy of the **present model** for laminar combustion rates, we compared the calculated laminar combustion rates for $\text{NH}_3/\text{H}_2/\text{Air}$ and $\text{NH}_3/\text{CH}_4/\text{Air}$ flames with experimental values reported in the literature. Furthermore, to identify the dominant reaction governing the laminar combustion velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flames, we calculated and compared the concentration distributions of chemical species within the computational domain and sensitivity analyses with respect to the mole fraction of chemical species using **present mech** and **Okafor Reduced Mech** [10]. Furthermore, by comparing the reaction rate constants between the **present mechanism** and the **Okafor Reduced Mechanism** [10], we quantitatively discussed how the improvements made to construct the **present mechanism** contribute physically. From these results, we identified the key reactions in the $\text{NH}_3/\text{H}_2/\text{Air}$ flame.

2.1 Calculation Conditions

Table 2-1 Chemkin Presetting

Equivalence ratio	Initial pressure [atm]	Initial temperature [K]
1.0	1.0	298.0

Table 2-1 shows the initial conditions for Chemkin, and Table 2-2 shows the calculation conditions for xH_2 and xCH_4 . All calculations were performed at an equivalence ratio of 1.0, an initial pressure of 1.0 atm, and an initial temperature of 298.0 K. Hereinafter, xH_2 represents the hydrogen content in the fuel, and xCH_4 represents the methane content in the fuel. In $NH_3/H_2/Air$ combustion, the characteristics vary depending on the hydrogen content in the fuel. In this study, the hydrogen content in the fuel is defined according to Equation (2-1). Furthermore, the characteristics of $NH_3/CH_4/Air$ combustion vary depending on the hydrogen content in the fuel. In this study, the hydrogen content in the fuel is defined according to Equation (2-2).

Q_{H_2} , Q_{NH_3} , and Q_{CH_4} represent the molar fractions of hydrogen, ammonia, and methane in the fuel, respectively.

$$xH_2 = \frac{Q_{H_2}}{Q_{H_2} + Q_{NH_3}} \quad (2-1)$$

$$xCH_4 = \frac{Q_{CH_4}}{Q_{CH_4} + Q_{NH_3}} \quad (2-2)$$

Table 2-2: xH_2 and xCH_4 : Computational Conditions

xH_2	0	0.20	0.50	0.80
xCH_4	0	0.20	0.40	0.60

In this study, for $NH_3/H_2/Air$ combustion, calculations were performed using the following parameters: xH_2 0.00, 0.20, 0.50, 0.80. Additionally, for $NH_3/CH_4/Air$ combustion, calculations were performed using the following parameters: xCH_4 0.00, 0.20, 0.40, 0.60. In all cases, the computational domain size was 10 cm. For the $NH_3/H_2/Air$ and $NH_3/CH_4/Air$ combustion calculations, automatic estimation of the temperature profile was employed. On the other hand, for the NH_3/Air combustion calculation, “Use Multicomponent Transport” was employed to improve the convergence of the calculation.

2.2 Theory and Governing Equations

In this study, steady-state, isobaric, quasi-one-dimensional laminar premixed flame simulations were performed using **Chemkin** [14–15] to conduct a detailed analysis of the structure and combustion characteristics of laminar premixed flames. By integrating chemical reaction mechanisms with advanced transport property models, this model enables the accurate calculation of temperature distributions within the flame zone and the spatial distributions of chemical species concentrations, including various radicals. In particular, for the prediction of combustion rates, the inclusion of physical factors such as thermal diffusion effects and radiative losses allows for the attainment of highly reliable calculation results under a wide range of temperature and pressure conditions.

The governing equations underlying this model and the definitions of each physical property are described below. The definitions of each variable are given in Table 2-7.

Table 2-7 Nomenclature

$\dot{m}$	Mass flow rate	$\text{g}/(\text{cm}^2\text{s})$
ρ	Density	g/cm^3
λ	Thermal conductivity	$\text{erg}/(\text{cm}\cdot\text{K}\cdot\text{s})$
u	Flow velocity	cm/s
A	Cross-sectional area	cm^2
T	Temperature	K
C_p	Specific heat of the mixture at constant pressure	$\text{erg}/(\text{g}\cdot\text{K})$
C_{pk}	Specific heat at constant pressure of the k th component	$\text{erg}/(\text{g}\cdot\text{K})$
V_k	Diffusion coefficient of the k th component	cm^2/s
D_{km}	Mixing-averaged diffusion coefficient	cm^2/s
D_{kj}	Multicomponent diffusion coefficient	cm^2/s
D_k^T	Thermal diffusivity (Sole effect)	$\text{g}/(\text{cm}\cdot\text{s})$
Y_k	Mass fraction of the k th component	-
X_k	Molar fraction of species k	-
ω_k	Reaction rate of species k	$\text{mol}/(\text{cm}^3\cdot\text{s})$
W_k	Molecular weight of species k	g/mol
$\bar{W}$	Average molecular weight of the mixture	g/mol
h_k	Specific enthalpy of the k th species	erg/g
K	Total number of chemical species	-
x	Spatial coordinates	cm

<i>t</i>	Time	s
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2.2.1 Flame-speed calculation model

used in this study The **flame-speed calculation** model is used to simulate **adiabatic, freely propagating** flames at specified pressure and inlet temperatures and to determine the laminar combustion velocity specific to the mixture. To describe steady, isobaric, quasi-one-dimensional flame propagation, the following governing equations are solved.

● Equations of Continuity

The **mass flow rate** $\dot{m}$ is constant and independent of the spatial coordinate x , and is expressed by Equation (2-3).

$$\dot{m} = \rho u A = \text{constant} \quad (2-3)$$

In standard laminar combustion velocity calculations, $A = 0.1$ (constant) is applied.

● Energy Conservation Equation

In this model, by definition, there is no heat loss to the exterior ($Q_{rad} = 0$). Since the laminar combustion rate depends on heat transport, the prediction of temperature distributions derived from the energy conservation equation shown in Equation (2-4) is essential for combustion rate calculations. Equation (2-4) integrates heat diffusion due to thermal conduction in the second term, heat transfer associated with species-specific diffusion in the third term, heat absorption and release associated with reactions in the fourth term, and radiative heat loss to the exterior in the fifth term.

$$\dot{m} \frac{dT}{dx} - \frac{1}{C} \frac{d}{dx} \left(A \lambda \frac{dT}{dx} \right) + \frac{A}{C} \sum_{k=1}^K \rho Y_k V_k C_{p,k} \frac{dT}{dx} - \frac{A}{C} \sum_{k=1}^K h_k \omega_k W_k + \frac{A}{C} Q_{rad} = 0 \quad (2-4)$$

● Equation for Conservation of Chemical Species

Equation (2-5) describes the spatial variation in the mass fraction of each chemical species k . The first term represents transport due to flow (convection), the second term represents diffusion due to the species-specific diffusion coefficient V_k , and the third term represents formation or destruction due to chemical reactions. A steady state is maintained when these terms are in equilibrium.

$$\dot{m} \frac{dY_k}{dx} + \frac{d}{dx} \left(A \rho Y_k V_k \right) - A \omega_k W_k = 0 \quad (k = 1, \dots, K) \quad (2-5)$$

● Modified Arrhenius equation

The net reaction **rate** ω_k for various chemical reactions is determined based on the law of mass action as the result of competition among all involved chemical reactions. The forward reaction rate constant k_f for each reaction is defined by the modified Arrhenius equation in Equation (2-6). This demonstrates that the reaction rate varies exponentially with temperature

. The characteristics of each reaction are defined by three parameters: the frequency factor A , the temperature exponent β , and the activation energy E_c .

$$k_f = AT^\beta \exp\left(-\frac{E_c}{RT}\right) \quad (2-6)$$

- **Equation of State for an Ideal Gas**
Equation (2-7) treats the mixture as an ideal gas and defines the relationship between mechanical variables (pressure, density) and thermodynamic variables (temperature) using the average molecular weight $\bar{W}$.

$$\rho = \frac{P\bar{W}}{RT} \quad (2-7)$$

2.2.2 Mixture-Averaged Transport

In this study, a mixture-averaged approximation model is adopted to accurately describe the transport phenomena of each chemical species in a multicomponent mixture while minimizing computational load. The equation shown below describes a model for simplifying and calculating the "motion" of each chemical species in a multicomponent mixture.

- **Definition of Diffusion Rate**
Equation (2-8) shows that the diffusion velocity of each species, V_k , consists of three components: normal diffusion, thermal diffusion, and the correction velocity. The total velocity at which each species moves, V_k , is the property of moving toward areas of lower concentration
We consider this in terms of $\bar{V}_k$, the property that follows the temperature distribution T_k , and the correction V_c , which is used to maintain computational consistency.

$$V_k = \bar{V}_k + T_k + V_c \quad (2-8)$$

- **Mixed-average diffusion coefficient**
Although diffusion in multicomponent systems is inherently complex, here it is approximated as the diffusion coefficient between a specific species k and the "rest of the mixture." Using the binary (two-component) diffusion coefficient D_{jk} , the effective diffusion coefficient of species k in the mixture is defined.

$$D_{km} = \frac{1 - Y_k}{\sum_{j \neq k}^K X_j / D_{jk}} \quad (2-9)$$

- Thermal diffusion rate

This considers diffusion caused by temperature gradients. The sign of the thermal diffusion coefficient Θ_k represents phenomena such as the movement of light molecules, like hydrogen, from lower-temperature regions to higher-temperature regions. This is considered only for light species such as light hydrogen (H), hydrogen molecules (H_2), and helium (He).

$$T_k = \frac{D_{km} \Theta_k}{X_k} \frac{1}{T} \frac{dT}{dx} \quad (2-10)$$

- Correction factor for mass conservation

Equation (2-11) is a correction term introduced to ensure that the sum of the mass flow rates of all species equals 0, i.e., to satisfy the law of mass conservation. When using an approximate model (the mixed-average method), simply summing the diffusion rates of each component may not result in exact mass conservation for the system as a whole. Equation (2-11) serves as a constraint to ensure that the total mass balance is strictly satisfied.

$$\sum_{k=1}^K Y_k V_k = 0 \quad (2-11)$$

2.2.3 Multicomponent Transport Properties

In this study, a multicomponent transport model is provided as an option for cases requiring a more accurate description of transport phenomena. This model is based on the Dixon-Lewis method and calculates multicomponent diffusion coefficients, thermal conductivities, and thermal diffusivities rigorously by solving a system of equations that includes two-component diffusion coefficients, molar fractions, and various thermodynamic and molecular properties. Particularly under conditions where the thermal diffusion effect (Sole effect) dominates, this model demonstrates significantly higher accuracy than the mixed-average approximation model, making it suitable for precise analysis of flame structures.

- Definition of diffusion velocity

In a multi-component transport model, the diffusion velocity V_k is expressed as the sum of the normal diffusion velocity $\bar{V}_k$ due to the concentration gradient and the thermal diffusion velocity T_k due to the temperature gradient. In a multi-component model, the sum of the diffusion fluxes for each component is theoretically zero; therefore, it is defined without introducing the correction velocity V' required in the mixed-average method.

$$V_k = \bar{V}_k + T_k \quad (2-12)$$

- Normal diffusion rate (multicomponent diffusion)

This demonstrates that the diffusion rate of a given species k is determined by the matrix of relative diffusion coefficients D_{kj} between that species and all other species j in the mixture. This provides a rigorous evaluation of the interference effects between multiple components that cannot be fully captured by the simple Fick's law (mixing-average method). By using the multi-component diffusion coefficient D_{kj} , the exact diffusion rate that accounts for the interactions among all chemical species can be calculated.

$$\vec{V}_k = \frac{1}{X_k \vec{W}} \sum_{j=1}^K W_j D_{kj} \frac{dX_j}{dx} \quad (2 - 13)$$

- Thermal diffusion coefficient

This represents mass transport (the Soret effect) that occurs in proportion to the temperature gradient ∇T . In multi-component transport models, this formulation provides significantly higher computational accuracy than the mixed-average approximation in analyses where the effects of thermal diffusion are significant, such as in hydrogen combustion.

$$T_k = - \frac{D_k^T}{\rho Y_k T} \nabla T \quad (2 - 14)$$

2.3 Reaction Mechanism

The objective of this study is to develop a reaction mechanism capable of accurately predicting laminar combustion rates under both NH₃/CH₄/Air and NH₃/H₂/Air flame conditions, with a particular focus on identifying the key combustion mechanisms in NH₃/H₂/Air flames. In the course of this investigation, four types of reaction mechanisms were employed: the Nakamura mechanism [16], the Otomo mechanism [17], the Okafor reduced mechanism [10], and the Gotama detailed mechanism [11].

In this section, we describe the objectives and characteristics of each reaction mechanism, as well as their role in this study.

1. Nakamura Mech

Nakamura Mech (38 species, 232 reactions) is a reaction mechanism developed to comprehensively predict ammonia oxidation from low to high temperatures. This mechanism was constructed based on the mechanisms of Mathieu and Petersen and Konnov et al., with significant updates to the N₂H_x-related reaction pathways. It has been confirmed that the mechanism accurately reproduces the concentration profiles of NH₃, O₂, and H₂O in the stoichiometric range of 0.8 to 1.2 under atmospheric pressure. Furthermore, it has been validated in both the temperature range of weak flames (approximately 1300 K) and the high-temperature range of shock tube experiments (approximately 2000 K). Sensitivity analysis and reaction pathway analysis have identified the importance of N₂H chemistry, particularly in the low-temperature region around 1300 K, and this contributes to the prediction of chemical species profiles within weak flames.

. This mechanism accurately predicts the final molar fractions of NO and N₂O downstream of the reaction zone and shows reasonable agreement with the ignition delay time and laminar combustion velocity reported in the literature. On the other hand, challenges remain, such as the overestimation of NO and N₂O within the reaction zone of weak flames, and the need for further refinement of the chemical reaction kinetics related to the ammonia ignition process has been pointed out [16].

2. Otomo Mech

Otomo Mech (32 species, 213 reactions) is a reaction mechanism developed to comprehensively predict NH₃/air flames in terms of both laminar combustion velocity and ignition delay time. This mechanism was constructed based on prior research by Song et al., through the scrutiny and refinement of the reaction rate constants of related elementary reactions, including NH₂, HNO, and N₂H₂. It has been demonstrated that re-examining the reactivity of these chemical species is crucial for improving the predictive accuracy of ammonia oxidation. Furthermore, a key feature of this model is its ability to clearly explain reported ignition delay times and laminar combustion rates under a wide range of equivalence ratios and pressure conditions. In particular, it has been confirmed that the calculated ignition delay times agree very well with experimental data. Specifically, in the verification of NH₃/H₂/Air flames under conditions ranging from 1 to 5 atm, this model predicts experimental data well, particularly under high pressures such as 3 atm and 5 atm. This is attributed to the fact that the improved NH₃ combustion reaction mechanism acts dominantly in the region where the molar fraction of H₂ is low (< 0.2). On the other hand, it has been reported that in regions where the hydrogen mixing ratio is high, H₂ combustion becomes dominant, and consequently, the difference in prediction accuracy compared to conventional models tends to be relatively small [17].

3. Okafor Reduced Mech

The **Okafor Reduced Mechanism** (42 species, 130 reactions), adopted as the basis for this study, is a reaction mechanism developed by Okafor et al. to model NH₃/CH₄/Air and NH₃/Air flames. This mechanism was developed with the aim of reducing the computational load in simulations of NH₃/CH₄/Air flames. Reaction rate constants have been optimized based on extensive measurement data to enable accurate prediction of laminar combustion rates while reducing the number of reactions compared to the parent model, the **Okafor Detailed Mech**. Specifically, it has been validated using measurement results of laminar combustion rates and Machstein lengths under high-pressure conditions up to 0.5 MPa, confirming its high predictive accuracy. This mechanism not only faithfully reproduces the laminar combustion rate of NH₃/CH₄/Air- s within the range of experimental uncertainty but also exhibits good predictive accuracy for NH₃ consumption characteristics and the formation behavior of NO and CO. In particular, it has been clarified that the NO concentration in NH₃ flames strongly depends on the O and H radical pools, and the predictive accuracy of fuel NO chemistry has been improved through appropriate modeling of rate-limiting reactions [10].

In this study, we adopted this mechanism—which exhibits extremely high reliability in NH₃/CH₄/Air flames—as the basis for our model and proposes modifications to improve the accuracy of laminar combustion rate predictions in NH₃/H₂/Air flames

. The objective is to identify the combustion mechanisms that play a dominant role, particularly in NH₃/H₂/Air flames.

4. Gotama Detailed Mech

The **Gotama Detailed Reaction Mechanism** (32 types, 165 reactions) is a re-mechanism developed to accurately predict the combustion characteristics of NH₃/H₂/Air flames, particularly under **high-pressure** and rich conditions. Existing reaction mechanisms (Han et al., 2019)[13] have been noted to underestimate combustion in NH₃/H₂/Air flames under low-pressure and lean conditions, and to overestimate it under high-pressure and rich conditions. To resolve these discrepancies, this mechanism focuses on the interactions between the oxidation pathways of NH₃ and H₂ and aims to optimize the chemical reaction processes. The model has been validated using measured laminar combustion rates under a wide range of fuel-rich conditions—up to pressures of 0.5 MPa and equivalence ratios of 1.8—for which data were scarce in previous studies, and high reliability has been reported. Furthermore, to reduce computational load, the model has been simplified from a detailed mechanism to 26 species 119 reactions.

(**Reduced Mech**) is also being conducted. Furthermore, the sensitivity to flame propagation, as well as to the combustion rate

Emphasis is also placed on predicting the **Markstein** length. By accurately reproducing the equivalence ratio dependence of the Markstein length under high pressure, it enables the analysis of flame instability caused by hydrodynamic and thermal diffusion effects. Through reaction analysis, the study elucidates the influence of the "**H₂ recovery mechanism**," in which NH₃-derived species are involved, on the **H** radical pool. In particular, it has been demonstrated that, in the intermediate hydrogen mixing ratio range, reaction systems involving **N₂H_x** kinetics significantly contribute to the behavior of **H** and **H₂**. By refining the kinetics of these reactions, improved prediction accuracy has been achieved [11].

3. Results and Discussion

3.1 Verification of the Prediction Accuracy of Laminar Flame Speeds for Reaction Mechanisms

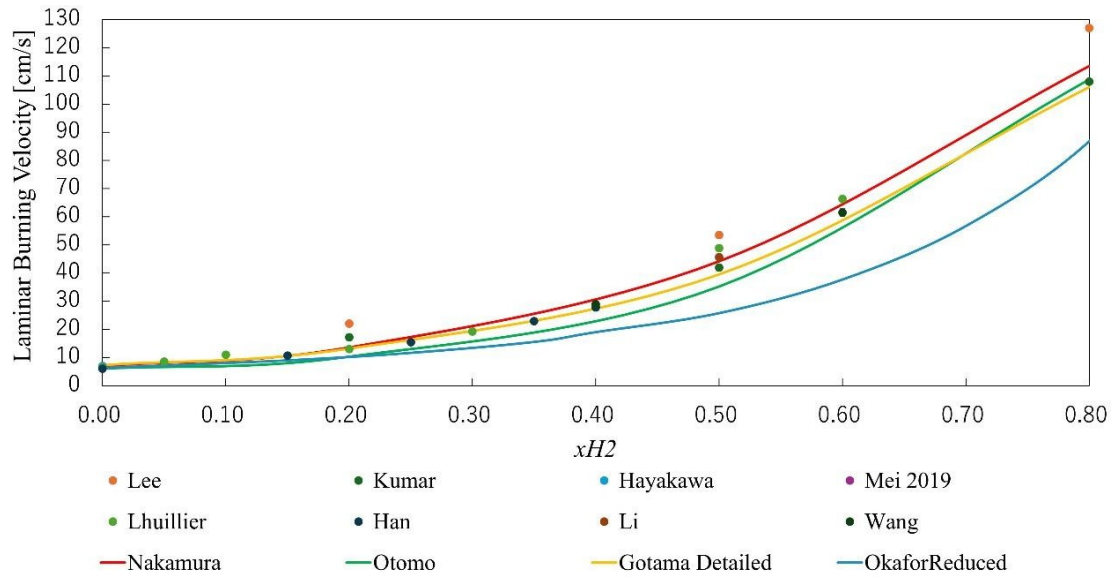


Figure 3-1 Experimental and calculated laminar flame speeds of NH₃/H₂/air flames under stoichiometric conditions as a function of xH_2

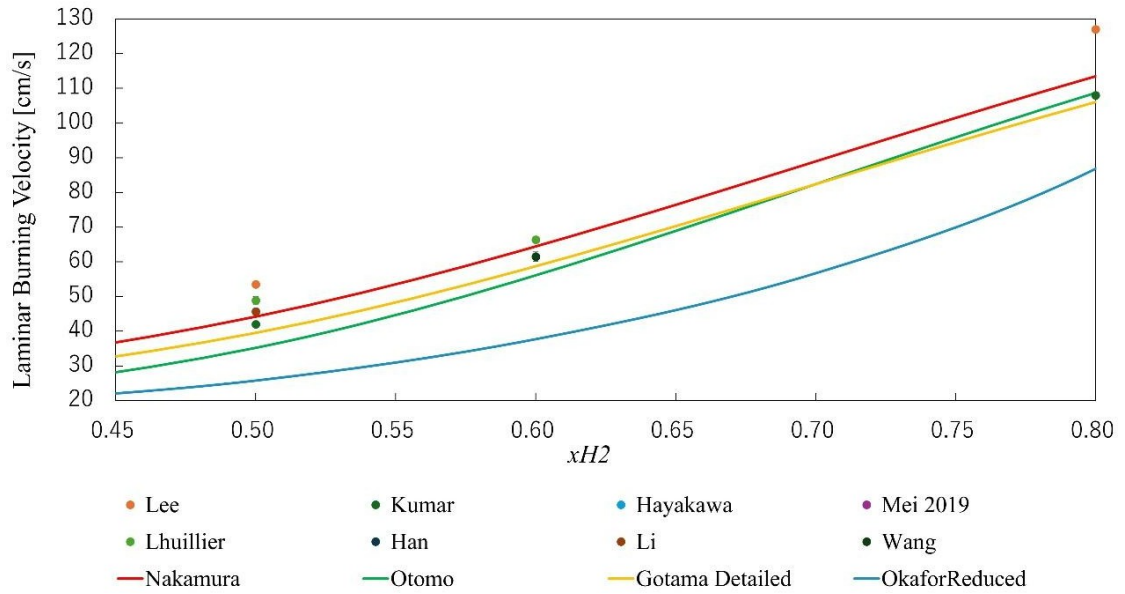


Figure 3-2 Experimental and calculated laminar flame speeds of NH₃/H₂/Air flames under stoichiometric conditions as a function of xH₂ (xH₂ = 0–0.55)

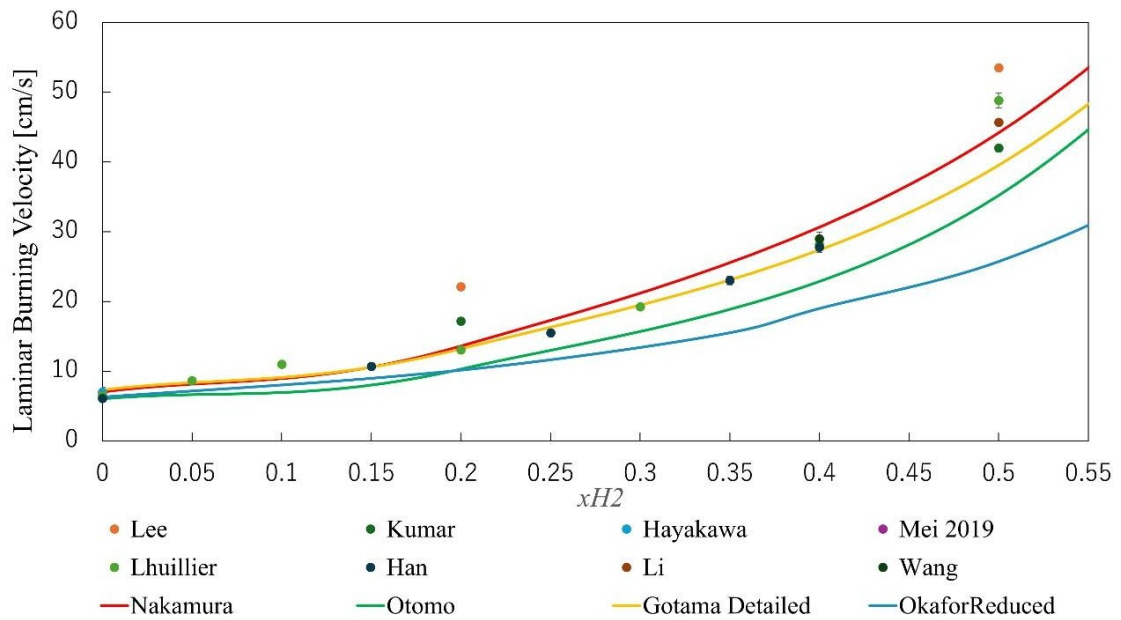


Figure 3-3 Experimental and calculated laminar flame speeds of NH₃/H₂/Air flames under stoichiometric conditions as a function of xH₂ (xH₂ =0.45–0.80)

Figures 3-1 to 3-3 show the calculated values from this study and literature values from other studies [4, 13, 18, 20–23] for the laminar combustion rate as a function of the hydrogen fraction under stoichiometric conditions for an $\text{NH}_3/\text{H}_2/\text{Air}$ flame. As shown in Figure 3-1, the calculated laminar combustion velocity increases nonlinearly as x_{H_2} increases, and the data from this study are consistent with the trends observed in the literature. Furthermore, as shown in Figures 3-1 to 3-3, the **Okafor Reduced Mech** model is able to accurately predict the laminar combustion velocity when x_{H_2} is 0—that is, for a pure NH_3/Air flame—but

It can be seen that as x_{H_2} increases, the laminar combustion rate is underestimated. Furthermore, while the **Otomo mechanism** accurately predicts the laminar combustion rate for NH_3/air flames, it slightly underestimates the laminar combustion rate as x_{H_2} increases. On the other hand, it is evident that both the **Gotama Gas-Phase-Kinetics Detailed Mech** and the **Nakamura Mech** accurately predict the laminar combustion rate for all hydrogen fractions (x_{H_2}).

To identify the cause of the underestimation in the **Okafor Reduced Mech**, we revisited the reaction characteristics of its parent model, the **Okafor Detailed Mech**, to gain insights into the underlying factors. The **Okafor Detailed Mech** was developed by Okafor et al. primarily for simulating $\text{NH}_3/\text{CH}_4/\text{Air}$ flames, and its structure is based on **GRI-Mech 3.0** [24] and incorporates key ammonia oxidation reactions from **Tian Mech** [25]. Han et al. [13] reported that **Okafor Detailed Mech** predicts laminar combustion rates very accurately for both CH_4/Air and NH_3/Air flames, regardless of the equivalence ratio, and that it predicts rates more accurately for CH_4/Air flames. On the other hand, they noted that while the **Okafor Detailed Mechanism** predicts laminar combustion rates for $\text{NH}_3/\text{H}_2/\text{Air}$ flames up to $x_{\text{H}_2} \approx 0.15$, it underestimates laminar combustion rates when $x_{\text{H}_2} > 0.15$. Han et al. noted that, given the fact that the **Okafor Detailed Mech** accurately predicts CH_4/Air flames ($\text{H}/\text{O}/\text{C}$ system), the primary cause of the underestimation of laminar combustion rates in $\text{NH}_3/\text{H}_2/\text{Air}$ flames is likely not the $\text{H}/\text{O}/\text{C}$ submechanism itself, but rather the inaccuracy of the reaction rate constants involving nitrogen-containing species.

In this study, the underestimation observed in the **Okafor Reduced Mechanism** is also attributed to the reaction rate constants involving nitrogen-containing species.

We hypothesized that this was due to inaccuracies in the rate constants. In particular, in $\text{NH}_3/\text{H}_2/\text{Air}$ flames, compared to $\text{NH}_3/\text{CH}_4/\text{Air}$ flames, H and OH radicals are abundant in the radical pool; therefore, we speculate that the formation of NH_x species accompanying the dehydrogenation of NH_3 is promoted, leading to a significant increase in the concentrations of NH_2 and NH . Consequently, recombination reactions that generate N_2H_x species such as N_2H_2 are promoted [16], and furthermore, as pointed out by Gotama et al. [11], the importance of the pathway via NNH , which has a significant impact on the H radical pool, is expected to increase. Based on the above background, a reaction pathway analysis was performed to identify the dominant factors causing the underestimation of laminar combustion rates in the $\text{NH}_3/\text{H}_2/\text{Air}$ flame of the **Okafor Reduced Model**.

3.2 Characterization and Comparison of Reaction Mechanisms

By analyzing the characteristics of each reaction mechanism and comparing those of the **Okafor Reduced Mechanism** with other reaction mechanisms, we identified the specific cause of the **Okafor Reduced Mechanism's** underestimation of the laminar combustion rate in $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

We attempted to identify the dominant factors. Specifically, by comparing the reaction pathways of **Gotama Detailed Mech**, and **Nakamura Mech**—which accurately predicted the laminar combustion rate—with those of **Okafor Reduced Mech**, we sought to identify the reactions and chemical species responsible for the underestimation in **Okafor Reduced Mech**. Furthermore, to clarify the influence of hydrogen on the combustion characteristics of the **NH₃/H₂/Air** flames, this study focused exclusively on the conditions $\phi = 1.0$. Additionally, to concisely discuss the fundamental differences in reaction mechanisms, the analysis was limited to stoichiometric conditions ($\phi = 1.0$).

To characterize the contribution of each reaction to the laminar combustion rate of an **NH₃/H₂/Air** flame, the integral of the rate of production (IROP) was evaluated, and reaction pathway diagrams were constructed for the major chemical species related to **NH₃** and **H₂** in the fuel. Figures 3-4, 3-5, and 3-6 show the reaction pathway diagrams obtained from the calculation results of the **Okafor Reduced Mech**, **Nakamura Mech**, and **Gotama Detailed Mech** models under stoichiometric conditions for the **NH₃/H₂/Air** flame ($\phi = 1.0$). Here, IROP was calculated using Equation (3-1).

$$I_{ROP} = \int_0^L \dot{\omega}_{R,i} dx \quad (3-1)$$

In Equation (3-1), $\dot{\omega}_{R,i}$ is the production rate of species i in elementary reaction R , and the upper limit L of the integration range was set to 10 cm, corresponding to the entire computational domain.

The criteria for drawing the reaction pathway diagrams shown in Figures 3-4 to 3-6 and the comparison methods are described below.

① Visualization and Normalization of Pathways

The thickness of the pathways (arrows) in each figure was normalized based on the value of the **OH→H₂O** pathway, which has the largest I_{ROP} among all pathways. The thickness of the pathway with the largest I_{ROP} was defined as 10 pt, and only pathways with a thickness of 1% or more relative to this maximum value were included. Some pathways were omitted to avoid complex intersections. Near each pathway, the relative thickness relative to the maximum value is noted as “[PPT: X.X pt].” Additionally, the value of the maximum I_{ROP} is shown at the top of the pathway diagram.

② Comparison of Reaction Mechanisms (Relative Size)

To evaluate the characteristics of each reaction mechanism relative to the **Okafor Reduced Mechanism**, we defined the ratio of I_{ROP} within the same pathway as “Relative Size.” For example, if the I_{ROP} of a specific pathway in the **Nakamura Mechanism** is twice that of the **Okafor Reduced Mechanism**, we denote the Relative Size as 200%, thereby clarifying the quantitative differences between the mechanisms.

③ Display of Contribution by Elementary Reactions

For the I_{ROP} of each pathway () and the I_{ROP} of the **Okafor Reduced Mechanism** (), we listed the elementary reactions with the largest contributions along with their reaction numbers (#) and the percentage

(%) are shown.

An example from the figure is shown below.

[PPT: 6.4pt] Relative Size: 136.1% #3:

OH+H2OH+H2O (68.9%)

#2: O+H2OH+OH (31.1%)

This is the notation for $\text{H}_2 \rightarrow \text{H}$ in Figure 3-6. The thickness of this pathway is denoted as “[PPT: 6.4pt],” indicating that it has an IROP size of 64% relative to *the* largest IROP (10.0 pt) in the entire reaction mechanism.

Since the **Relative Size** is 136.1%, it can be seen that the IROP of this pathway in the present reaction mechanism has increased by approximately 1.36 times compared to the same pathway in **the Okafor Reduced Mech** used as a reference. This suggests that the species conversion process of $\text{H}_2 \rightarrow \text{H}$ is more promoted in this mechanism.

A breakdown of the elementary reactions comprising this pathway shows that elementary reaction #3: $\text{OH} + \text{H}_2\text{O} \rightarrow \text{H}^\bullet + \text{H}_2\text{O}$ accounts for 68.9%—a majority—followed by elementary reaction #2: $\text{O} + \text{H}_2\text{O} \rightarrow \text{H}^\bullet + \text{OH}$, which contributes 31.1%.

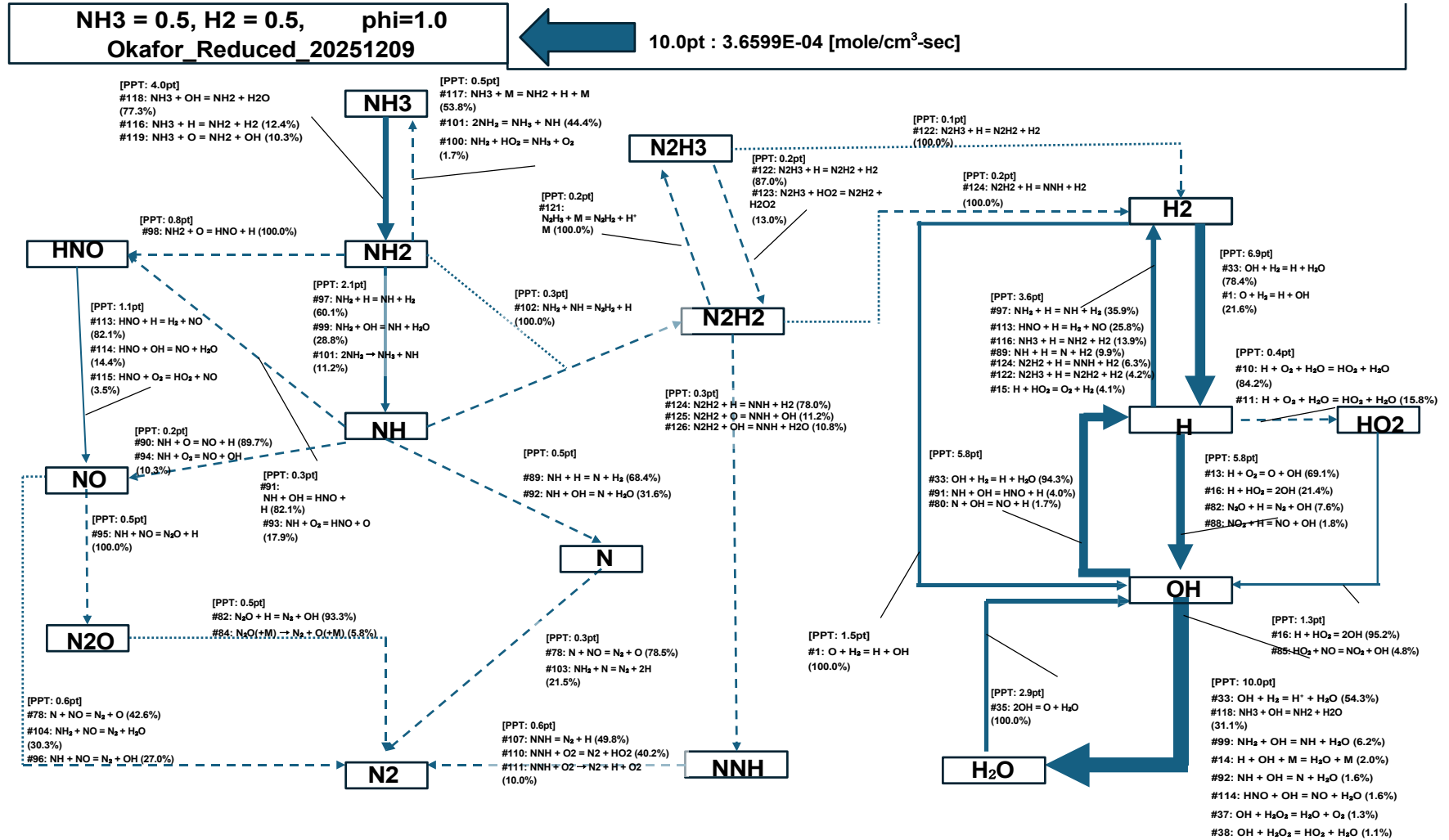


Figure 3-4 Reaction pathway of NH₃/H₂/air flames at $x\text{H}_2 = 0.50$ and $\phi = 1.0$ calculated using the Okafor Reduced Mechanism

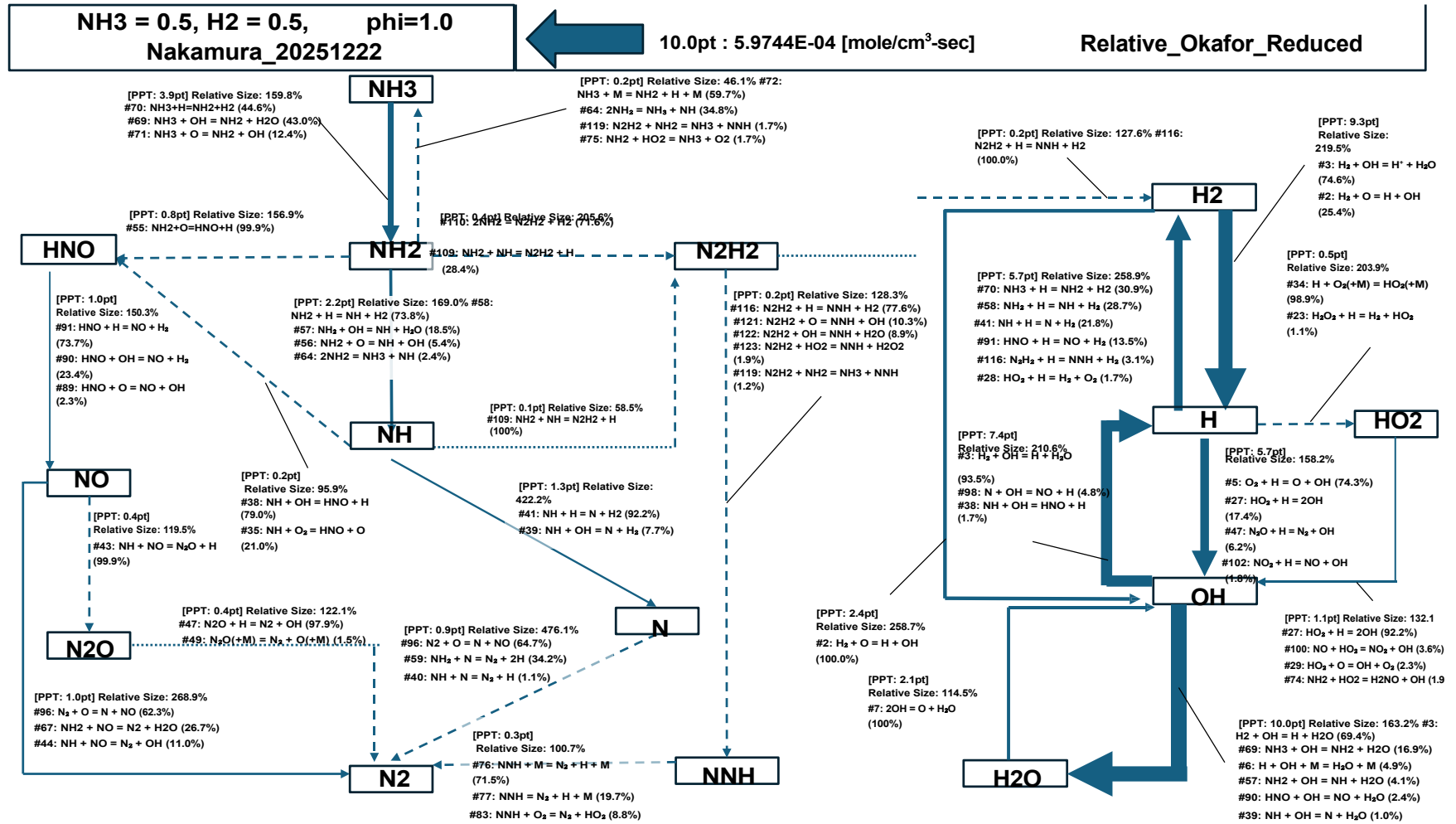


Figure 3-5 Reaction pathway of NH₃/H₂/air flames at $x_{H_2} = 0.50$ and $\phi = 1.0$ calculated using the Nakamura mechanism

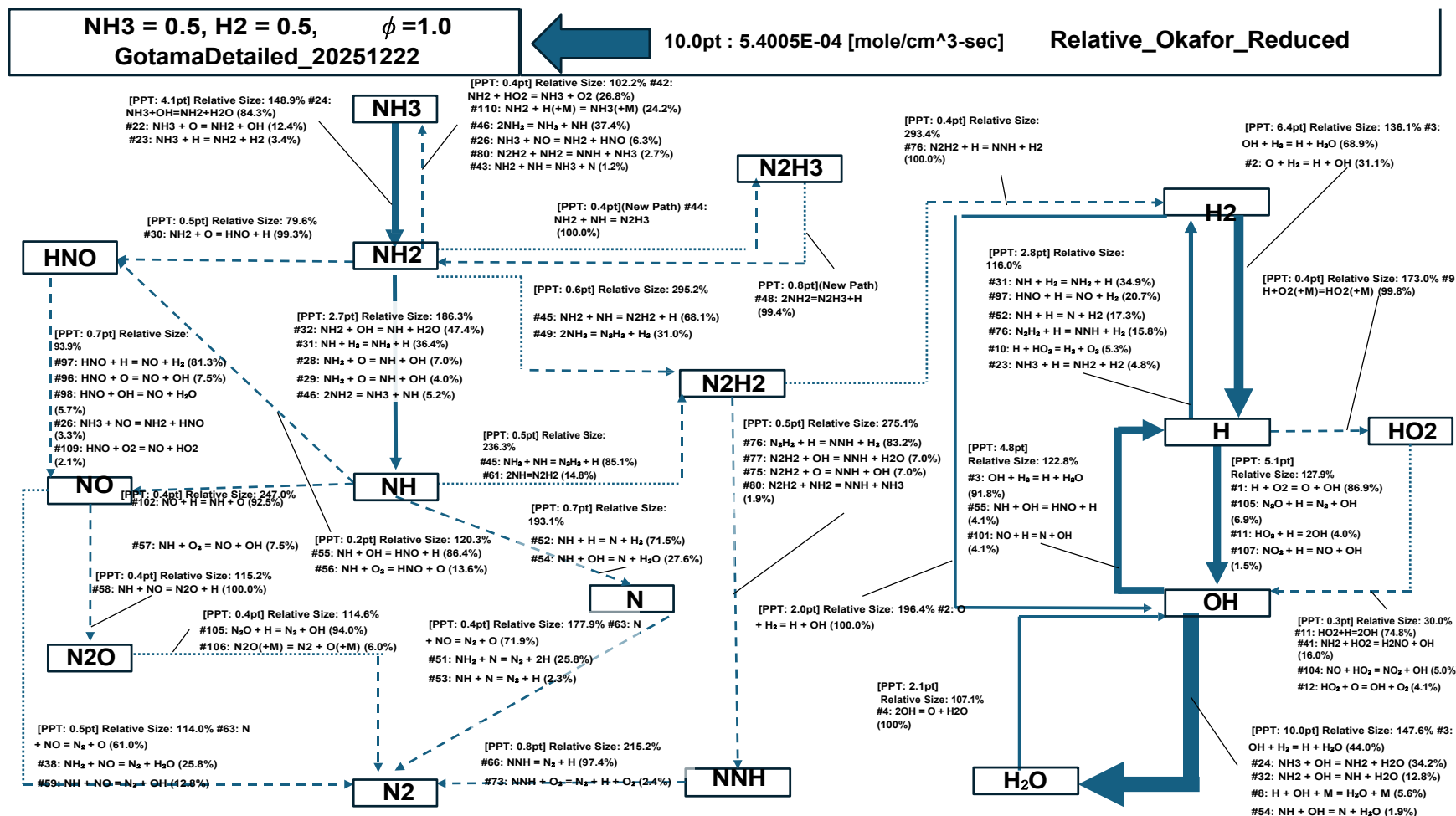


Figure 3-6 Reaction pathway of NH₃/H₂/air flames at $x_{H_2} \phi = 0.50$ and $\phi = 1.0$ calculated using the Gotama Detailed Mechanism

3.2.1 Comparison and Analysis of the Characteristics of the Okafor Reduced Mechanism and the Nakamura Mechanism

To identify the dominant factors causing the **Okafor Reduced Mech** to underestimate the laminar combustion rate of **NH₃/H₂/Air** flames, we compared the reaction pathways of **the Okafor Reduced Mech** and **the Nakamura Mech**, revealing differences in the following four pathways.

① $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$

As shown in Figure 3-5, the I_{ROP} for the $\text{NH}_3 \rightarrow \text{NH}_2$ pathway in the **Nakamura mechanism** is 1.6 times higher than that for the same pathway in the **Okafor Reduced mechanism**. Furthermore, the I_{ROP} for the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2$ pathway has increased to approximately 2 times that of the same pathway in the **Okafor Reduced Mech**. On the other hand, looking at the I_{ROP} for the subsequent steps, $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$, the increases are limited to approximately 1.3 times and a similar magnitude, respectively, compared to **the Okafor Reduced Mech**. While the upstream reaction ($\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{N}_2\text{H}_2$) has been significantly enhanced, the small increase in the I_{ROP} for the downstream reaction suggests that, in the **Nakamura mechanism**, the $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ pathway does not have a significant influence on the laminar combustion rate of the **NH₃/H₂/Air** flame.

② $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N} \rightarrow \text{N}_2$

As shown in Figure 3-5, the I_{ROP} for the $\text{NH} \rightarrow \text{N} \rightarrow \text{N}_2$ pathway in the **Nakamura Mechanism** has increased significantly to approximately 4 times and 4.7 times, respectively, compared to the same pathway in the **Okafor Reduced Mechanism**. In particular, R3-1, which makes the largest contribution to the $\text{N} \rightarrow \text{N}_2$ pathway, also makes the largest contribution to the I_{ROP} in the $\text{NO} \rightarrow \text{N}_2$ pathway.



This is thought to be caused by the increased concentration of **H** radicals due to hydrogen co-combustion, which accelerates the $\text{NH} \rightarrow \text{N}$ reaction rate and thereby increases the **N** concentration.

③ $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO} \rightarrow \text{N}_2$

As shown in Figure 3-5, the I_{ROP} for the $\text{HNO} \rightarrow \text{NO} \rightarrow \text{N}_2$ pathway in the **Nakamura Mechanism** has increased to approximately 1.5 times and 2.7 times, respectively, compared to the same pathway in the **Okafor Reduced Mechanism**. In the $\text{NO} \rightarrow \text{N}_2$ pathway, the aforementioned R3-1 shows the largest contribution. On the other hand, R3-2 is dominant in the $\text{HNO} \rightarrow \text{NO}$ pathway. These results suggest that the increase in **H** radical concentration due to hydrogen co-combustion promotes **NO** production via R3-2, thereby significantly influencing the combustion characteristics of the **NH₃/H₂/Air** flame.



- ④ In the **Nakamura mechanism**, the importance of the pathway related to N_2H_3 decreases

As shown in Figure 3-5, in the **Nakamura Mech** model, the importance of the pathway involving N_2H_3 —which was identified in the **Okafor Reduced Mech** model (Figure 3-4)—has decreased, and it has been excluded from the pathway diagram. This suggests that, in the **Nakamura Mech** model, the influence of reactions via N_2H_x on the laminar combustion rate of $NH_3/H_2/Air$ flames is limited.

Analysis of points ① through ④ indicates that, while the contribution of fuel consumption pathways via N_2H_x decreases in the **Nakamura Mech**, the importance of pathways involving NO —particularly the $HNO \rightarrow NO \rightarrow N_2$ pathway—relatively increases. This is thought to be because hydrogen co-combustion increases the concentration of H radicals, which in turn promotes the formation of OH and O radicals, thereby activating the NO production and consumption cycle. Overall, this suggests that the prediction of laminar combustion rates in the **Nakamura Mech** model depends heavily on NO -related reactions and the behavior of O and OH radicals.

3.2.2 Comparison and Analysis of the Characteristics of the Okafor Reduced Mechanism and the Gotama Detailed Mechanism

To identify the dominant factors causing the **Okafor Reduced Mech** to underestimate the laminar combustion rate of $NH_3/H_2/Air$ flames, we compared the reaction pathways of the **Okafor Reduced Mech** and the **Gotama Detailed Mech**, and identified differences in the following three pathways.

- ① $NH_2 \leftrightarrow N_2H_3$

The $NH_2 \leftrightarrow N_2H_3$ pathway, which was not considered in the **Okafor Reduced Mechanism**, was identified as a significant pathway in the **Gotama Detailed Mechanism**. The reactions involving the formation and consumption of N_2H_3 are shown in R3-3 and R3-4.



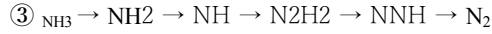
An examination of these reactions reveals that the $NH_3 \leftrightarrow N_2H_3$ and $NH_2 \leftrightarrow N_2H_3$ pathways directly contribute to changes in the concentrations of NH_2 and NH , which are key intermediates in the ammonia oxidation pathway. Therefore, the fact that this pathway is not included in the **Okafor-Reduced** and **Mech** suggests that this may lead to underestimates of the concentrations of major radicals and, consequently, of the laminar combustion rate.

- ② $NH_3 \rightarrow NH_2 \rightarrow N_2H_2 \rightarrow NNH \rightarrow N_2$

As shown in Figure 3-6, the $IROP$ values for the $NH_3 \rightarrow NH_2 \rightarrow NNH \rightarrow N_2$ pathway in the **Gotama Detailed Mechanism** are significantly higher than those for the same pathway in the **Okafor Reduced Mechanism**, at approximately 1.5 times, 3 times, 2.7 times, and 2.2 times, respectively. In particular, R3-5, which has the largest contribution (97.4%) in the $NNH \rightarrow N_2$ pathway, is a reaction that generates H radicals and is thought to contribute to the overall reaction promotion of the $NH_3/H_2/Air$ flame.



On the other hand, the contribution of R3-5 in the same pathway in the **Okafor Reduced Mech** model is only 49.8%. From this, it is highly likely that the underestimation of H radical generation by R3-5 lowers the overall reactivity of the flame, resulting in an underestimation of the laminar combustion rate of the NH₃/H₂/Air flame.



As shown in Figure 3-6, the IROP values for the $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$

pathways in the **Gotama Detailed Mechanism** are approximately 1.5 times, 1.9 times, and 2.4 times, approximately 2.8 times, and approximately 2.2 times, respectively, compared to the same pathway in the **Okafor Reduced Mechanism**. In particular, R3-6, which has the largest contribution (85.1%) in the $\text{NH}_2 \rightarrow \text{NH}$ pathway, is a reaction that generates H radicals and is likely one of the main reactions affecting the evaluation of the laminar combustion rate of NH₃/H₂/Air flames.



Given that the $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH}$ pathway in the **Gotama Detailed Mechanism** has a higher IROP than that in the **Okafor Reduced Mechanism**, it is highly likely that the reactivity of the ammonia consumption and decomposition processes involving R3-6 has been underestimated, leading to an underestimation of the laminar combustion rate of the NH₃/H₂/Air flame.

Based on the comparative analysis above, we believe that the primary factors causing the **Okafor Reduced Mech** model to underestimate the laminar combustion rate of NH₃/H₂/Air flames can be summarized as follows:

First, the omission and underestimation of N₂H_x-related pathways, such as the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2$ pathway. Consequently, we believe that the formation and consumption reactions of NH_2 and NH —important intermediates thought to influence the laminar combustion rate—have not been sufficiently accounted for.

Second, the contribution of reactions involving H radical generation, such as the $\text{NNH} \rightarrow \text{N}_2$ reaction (R3-5) and the $\text{NH}_2 \rightarrow \text{NH}$ reaction (R3-6), is low. It is thought that these factors act in combination to suppress the overall reactivity of the flame.

Furthermore, the reason we did not focus on updating the NO-related pathways based on the insights gained from the comparison with the **Nakamura Mechanism** in this study is that the **Okafor Reduced Mechanism**, which served as our baseline, has a proven track record of accurately predicting laminar combustion rates in oxygen-enriched combustion [26]. In oxygen-enriched combustion, large amounts of O and OH radicals are generated in the flame as the O₂ concentration increases; however, the **Okafor Reduced Mech** accurately predicted the laminar combustion rate of NH₃ in oxygen-enriched combustion. Therefore, we determined that modifying the NO-related pathways found in the **Nakamura Mech** was of low priority for improving the prediction accuracy of the laminar combustion rate in this study.

Based on the above considerations, and with reference to the **Gotama Detailed Mechanism**, we updated the series of xml-ph-0000@deeppl.internal N₂H_x xml-ph-0001@deeppl.internal formation pathways attributable to NH_x radicals in the **Okafor Reduced Mechanism**

, we aimed to construct a new reaction mechanism.

3.3 Newly constructed reaction mechanism

A comparison of the reaction pathways in **Gotama Detailed Mech** and **Okafor Reduced Mech** revealed that the clear difference between the two reaction mechanisms lies in the presence or absence of a series of N2Hx-related reactions originating from the NH2→NH pathway. In the **Gotama Detailed Mech**, N2Hx formation reactions starting from abundant NHx radicals are considered in detail, suggesting that these reactions play a dominant role in predicting laminar combustion rates. The specific mechanism obtained from the reaction pathway analysis of the **Gotama Detailed Mech** is as follows.

The N2Hx species generated by the recombination of abundant NHx radicals undergo a series of chain reactions and are ultimately converted to NNH. This NNH directly supplies H radicals to the reaction zone via the R3-7 pathway. Since H radicals activate chain reactions in the flame, they are thought to have a very significant effect on the laminar combustion rate, particularly in NH3/H2/Air flames rich in OH and H radicals.



Based on the above analysis, we have determined that the underestimation of the laminar combustion rate observed in the **Okafor Reduced Mechanism** is primarily due to the fact that the **Okafor Reduced Mechanism** fails to adequately describe the series of N2Hx formation reactions caused by the abundant NHx radicals characteristic of NH3/H2/Air flames.

Therefore, we extracted data related to NH₂, NNH, N₂H₂, N₂H₃, and N₂H₄ from the **Gotama Detailed- Mechanism** and incorporated them into the **Okafor Reduced Mechanism** to construct a new reaction mechanism. Hereinafter, the reaction mechanism proposed in this study is referred to as the “present mechanism.”

Table 3-1 lists the reactions added to the **Okafor Reduced Mechanism** and their rate constants. Table 3-2 lists the reactions in the **Okafor Reduced Mechanism** and their rate constants. Additionally, Table 3-3 presents the thermodynamic data and gas transport data for N₂H₄, a species added to the **Okafor Reduced Mechanism**. Furthermore, the reaction rate constants are expressed using the Arrhenius function in Equation (3-2). The units are cm, mol, s, cal, and K.

$$k = AT^n \exp\left(-\frac{E_a}{RT}\right) \quad (3-2)$$

Table 3-1 Reactions added to the Okafor Reduced Mech

Mechanism Index #	Reaction	A	n	Ea	Reference Citation (Source)
106	$\text{NH}_3 + \text{NH} + \text{M} = \text{N}_2\text{H}_4 + \text{M}$	1.09.E+15	0	0	C. Zetzsch, F. Stuhl (1981)
107	$\text{NH} + \text{NH} = \text{N}_2\text{H}_2$	6.26.E+13	-0.036	-161	S.J. Klippenstein, et al. (2009)
108	$\text{N}_2\text{H}_3 + \text{O} = \text{N}_2\text{H}_2 + \text{OH}$	1.70.E+08	1.5	-646	A.M. Dean, J.W. Bozzelli (2000)
110	$\text{N}_2\text{H}_3 + \text{OH} = \text{N}_2\text{H}_2 + \text{H}_2\text{O}$	1.20.E+06	2	-1192	A.M. Dean, J.W. Bozzelli (2000)
111	$\text{N}_2\text{H}_4 + \text{H} = \text{N}_2\text{H}_3 + \text{H}_2$	4.50.E+07	1.8	2613	D.-Y. Hwang, A.M. Mebel (2003)
112	$\text{N}_2\text{H}_4 + \text{OH} = \text{N}_2\text{H}_3 + \text{H}_2\text{O}$	4.00.E+13	0	0	P. Glarborg, et al. (1994)
114	$\text{NNH} + \text{O} = \text{N}_2 + \text{OH}$	1.20.E+13	0.145	-217	S.J. Klippenstein, et al. (2011)
117	$\text{NNH} + \text{OH} = \text{N}_2 + \text{H}_2\text{O}$	5.00.E+13	0	0	P. Glarborg, et al. (1998)
118	$\text{NNH} + \text{H} = \text{N}_2 + \text{H}_2$	1.00.E+12	0	0	J.W. Bozzelli, A.M. Dean (1995)
121	$\text{NNH} + \text{NO} = \text{N}_2 + \text{HNO}$	5.00.E+13	0	0	P. Glarborg, et al. (1998)
128	$\text{NH}_3 + \text{HO}_2 = \text{NH}_2 + \text{H}_2\text{O}_2$	3.00.E+11	0	22000	J.A. Miller, C.T. Bowman (1991)
129	$\text{NH}_3 + \text{NO} = \text{NH}_2 + \text{HNO}$	3.26.E+07	1.73	56600	
130, 131	$\text{NH}_2 + \text{O} = \text{NH} + \text{OH}$	7.00.E+12	0	0	
130, 131	$\text{NH}_2 + \text{O} = \text{NH} + \text{OH}$	8.60.E-01	4.01	1673	
135	$\text{NH}_2 + \text{O}_2 = \text{HNO} + \text{OH}$	2.90.E-02	3.764	18185	S.J. Klippenstein, et al. (2011)
140	$\text{NH}_2 + \text{NH} = \text{NH}_3 + \text{N}$	9.60.E+03	2.46	107	S.J. Klippenstein, et al. (2009)
141	$\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$	7.00.E+13	0	0	P.B. Pagsberg, et al. (1979)
144	$\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_3 + \text{H}$	1.20.E+12	-0.03	10084	A.M. Dean, J.W. Bozzelli (2000)
145	$\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$	7.83.E+11	0	0	N. Stothard, et al. (1995)
147	$\text{NH} + \text{NH} = \text{NH}_2 + \text{N}$	5.70.E-01	3.88	342	S.J. Klippenstein, et al. (2009)
148	$\text{N}_2\text{H}_2 + \text{NH} = \text{NNH} + \text{NH}_2$	2.40.E+06	2	-1192	A.M. Dean, J.W. Bozzelli (2000)
149	$\text{N}_2\text{H}_2 + \text{NO} = \text{N}_2\text{O} + \text{NH}_2$	4.00.E+12	0	11922	A.M. Dean, J.W. Bozzelli (2000)
150	$\text{N}_2\text{H}_2 + \text{NH}_2 = \text{NNH} + \text{NH}_3$	8.80.E-02	4.05	-1610	
151	$\text{N}_2\text{H}_3 + \text{O} = \text{NH}_2 + \text{HNO}$	3.00.E+13	0	0	A.M. Dean, J.W. Bozzelli (2000)
152	$\text{N}_2\text{H}_3 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{NH}_3$	9.20.E+05	1.94	-1152	A.M. Dean, J.W. Bozzelli (2000)

Table 3-2 Reactions updated in the Okafor Reduced Mechanism

Mechanism Index #	Reaction	A	n	Ea	Reference Citation (Source)
109	$\text{N}_2\text{H}_3 + \text{H} = \text{N}_2\text{H}_2 + \text{H}_2$	2.40.E+08	1.5	-10	A.M. Dean, J.W. Bozzelli (2000)
113	$\text{NNH} = \text{N}_2 + \text{H}$	1.00.E+09	0	0	S.J. Klippenstein, et al. (2011)
115	$\text{NNH} + \text{O} = \text{N}_2\text{O} + \text{H}$	1.00.E+14	0	0	P. Glarborg, et al. (1998)
116	$\text{NNH} + \text{O} = \text{NH} + \text{NO}$	3.30.E+14	-0.23	-1013	J.W. Bozzelli, A.M. Dean (1995)
119	$\text{NNH} + \text{O}_2 = \text{N}_2 + \text{HO}_2$	1.20.E+12	-0.34	149	J.W. Bozzelli, A.M. Dean (1995)
120	$\text{NNH} + \text{O}_2 = \text{N}_2 + \text{H} + \text{O}_2$	5.00.E+13	0	0	P. Glarborg, et al. (1998)
122	$\text{N}_2\text{H}_2 + \text{O} = \text{NNH} + \text{OH}$	3.30.E+08	1.5	497	A.M. Dean, J.W. Bozzelli (2000)
123	$\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$	1.10.E+14	0	3128	P. Glarborg, et al. (2018)
124	$\text{N}_2\text{H}_2 + \text{OH} = \text{NNH} + \text{H}_2\text{O}$	5.90.E+01	3.4	-1363	
125	$\text{NH}_3 + \text{O} = \text{NH}_2 + \text{OH}$	1.63.E+07	1.85	6457	D. Baulch, et al. (2005)
126	$\text{NH}_3 + \text{H} = \text{NH}_2 + \text{H}_2$	1.26.E+14	0	21500	R.K. Hanson, S. Salimian (1984)
127	$\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$	1.56.E+05	2.372	118.9	A. Stagni, et al. (2020)
132	$\text{HNO} + \text{H} = \text{NH}_2 + \text{O}$	3.50.E+15	-0.3	29200	N. Cohen, K.J. Westberg (1991)
133	$\text{NH} + \text{H}_2 = \text{NH}_2 + \text{H}$	2.10.E+13	0	15417	A. Fontijn, et al. (2006)
134	$\text{NH}_2 + \text{OH} \rightleftharpoons \text{NH} + \text{H}_2\text{O}$	9.60.E+06	1.97	669	S.H. Mousavipour, et al. (2009)
136	$\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$	1.60.E+16	-1.44	268	J. Park, M.C. Lin (1997)
137	$\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$	2.70.E+17	-1.85	0	W. Hack, et al. (1979)
138	$\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$	1.43.E+07	1.4	-1777	J. Park, M.C. Lin (1999)
139	$\text{NH}_2 + \text{HO}_2 = \text{NH}_3 + \text{O}_2$	4.50.E+13	0	0	O. Sarkisov, et al. (1984)
142	$\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$	4.30.E+14	-0.272	-77	S.J. Klippenstein, et al. (2009)
143	$\text{NH}_2 + \text{NH}_2 = \text{NH}_3 + \text{NH}$	5.60.E+00	3.53	552	S.J. Klippenstein, et al. (2009)
146	$\text{NH}_2 + \text{N} = \text{N}_2 + \text{H} + \text{H}$	5.00.E+13	0	0	K.P. Shrestha, et al. (2018)

Table 3-3 Chemical species data added to the Okafor Reduced Mech

[SPECIES]: N2H4											
Target Thermo Data:											
N2H4	HYDRAZINE	L	5/90N	2H	4	0	OG	200.000	6000.000	1000.	1
4.93957357E+00	8.75017187E-03	-2.99399058E-06	4.67278418E-10	-2.73068599E-14							2
9.28265548E+03	-2.69439772E+00	3.83472149E+00	-6.49129555E-04	3.76848463E-05							3
-5.00709182E-08	2.03362064E-11	1.00893925E+04	5.75272030E+00	1.14474575E+04							4
Target Transport Data:											
N2H4	2	205.000	4.230	0.000	4.260	1.500					

3.3.1 Verification of the Prediction Accuracy of Laminar Combustion Rates

To verify the validity of the present mechanism in NH₃/H₂/Air flames, the calculated laminar flame speeds from the present mechanism were compared with experimental values. Additionally, the calculated values were compared with those from the baseline Okafor Reduced Mechanism.

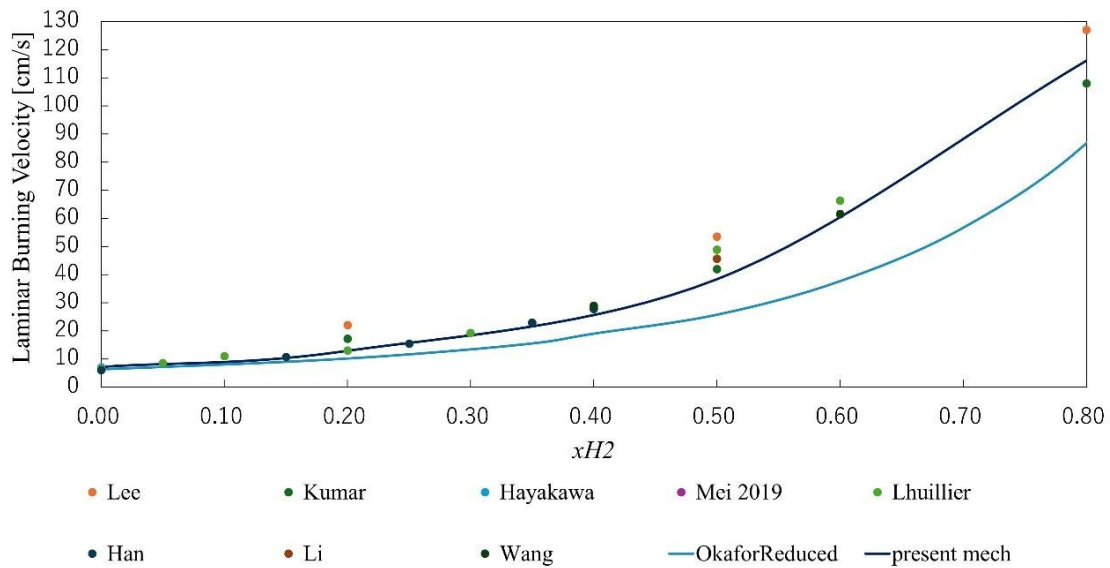


Figure 3-7 Experimental and calculated laminar flame speeds of NH₃/H₂/Air flames under stoichiometric conditions as a function of xH_2
(Comparison between the present and Okafor reduced mechanisms)

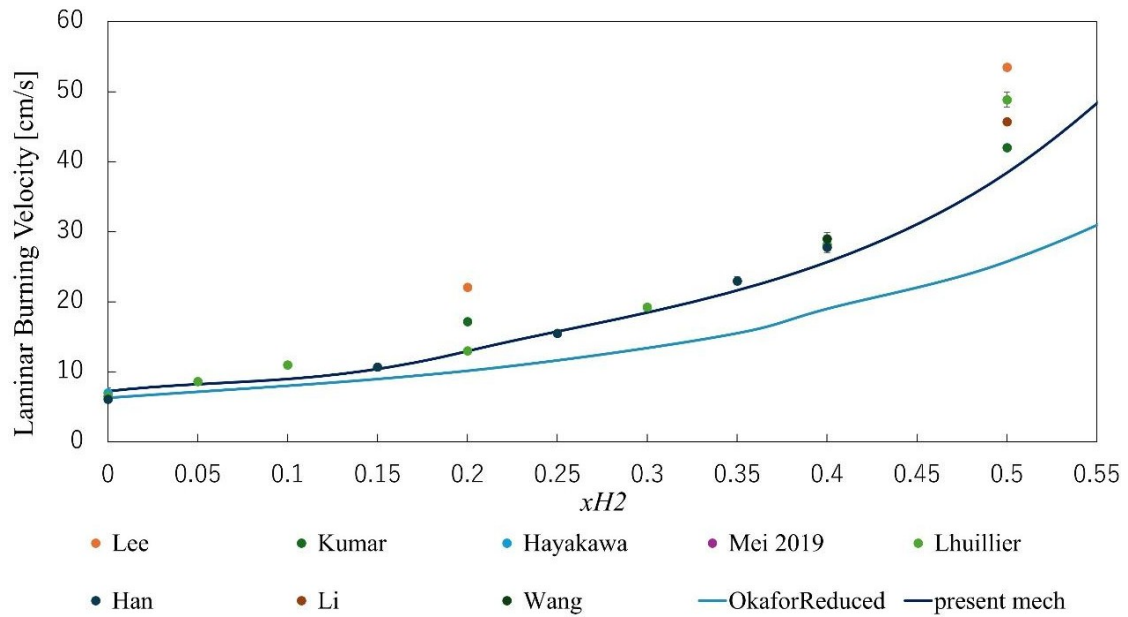


Figure 3-8 Experimental and calculated laminar flame speeds of NH₃/H₂/Air flames under stoichiometric conditions as a function of xH_2
(Comparison between the present and Okafor reduced mechanisms)
(xH_2 : 0–0.55)

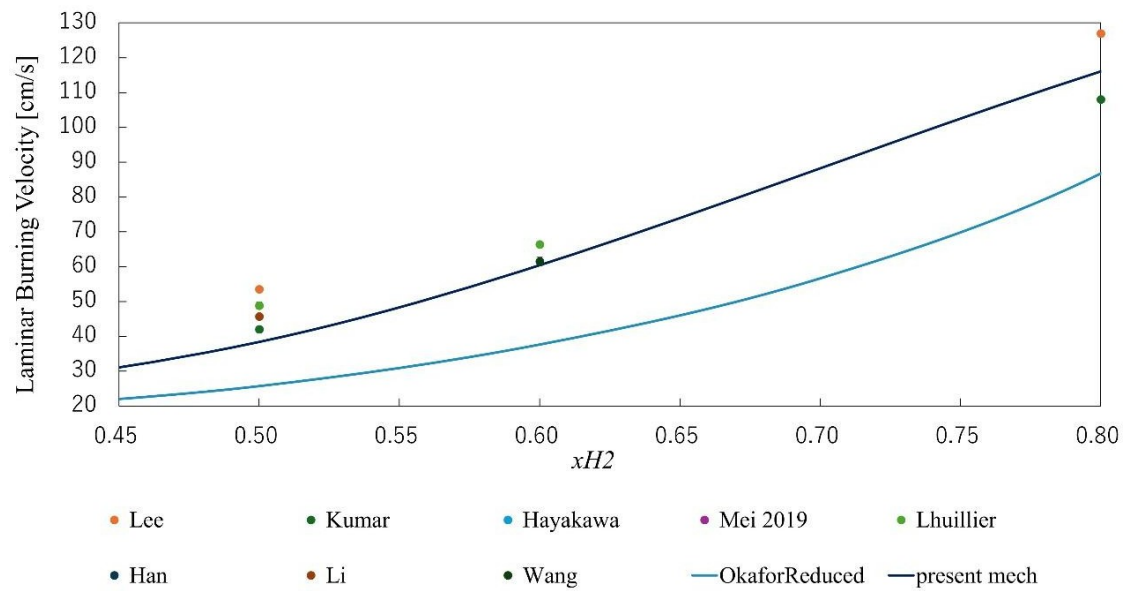


Figure 3-9 Experimental and calculated laminar flame speeds of NH₃/H₂/air flames under stoichiometric conditions as a function of xH_2
(Comparison between the present mechanism and the Okafor reduced mechanism)
(xH_2 : 0.45–0.80)

Figures 3-7 to 3-9 show the experimental values of laminar flame speeds for NH₃/H₂/Air flames under stoichiometric conditions as a function of the hydrogen fraction, along with the calculated values from the **present mechanism** and the **Okafor Reduced Mechanism**. As shown in Figures 3-7 to 3-9, it can be seen that the **present mechanism** adequately predicts the laminar flame speeds for all values of x_{H_2} . Furthermore, considering that the **Okafor Reduced Mechanism** was originally optimized for NH₃/CH₄/Air flames, we investigated whether the **present mechanism** can accurately predict the laminar combustion velocity for NH₃/CH₄/Air flames as well. Figure 3-10 shows the experimental values of the laminar combustion rate for x_{CH_4} under stoichiometric conditions for an NH₃/CH₄/Air flame, along with the calculated values from the **present mechanism** and the **Okafor Reduced Mechanism**.

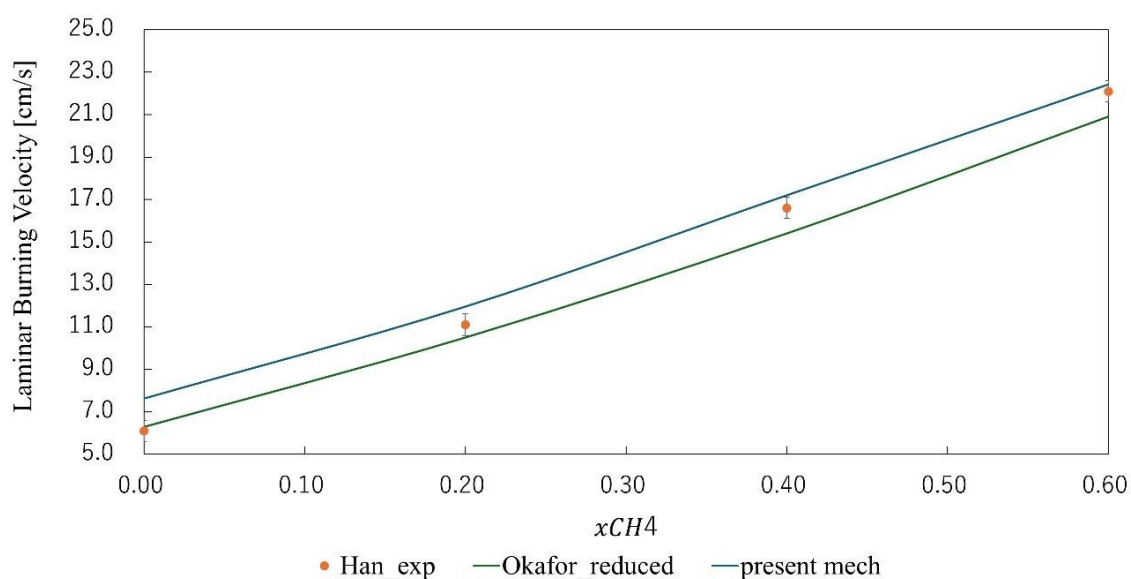


Figure 3-10 Experimental and calculated laminar flame speeds of NH₃/CH₄/Air flames under stoichiometric conditions plotted against the x_{CH_4} , ranging from 0 to 0.60

As shown in Figure 3-10, although the calculated values from the **present mechanism** are slightly higher than those from the **Okafor Reduced Mechanism**, it can be seen that they are able to predict values close to the experimental data. Furthermore, it can be seen that the predictions are accurate within the experimental error range, particularly under conditions where $x_{CH_4} > 0.40$.

Based on these results, it is considered that the **present model** successfully achieves both the ability to accurately predict the laminar combustion velocity of NH₃/H₂/Air flames and the preservation of the characteristic inherent to the **Okafor Reduced Model**, namely that it is “optimized for NH₃/CH₄/Air flames.”

As shown in Tables 3-1 through 3-3, it is evident that numerous reactions and data have been updated in the **present model**. However, it is difficult to identify from these lists which of the updated individual reactions plays a dominant role in the laminar combustion rate of the NH₃/H₂/Air flame. Therefore, we performed a sensitivity analysis on the laminar combustion rate of the NH₃/H₂/Air flame to identify the key submechanisms underlying its combustion characteristics.

3.4 Analysis of the Combustion Characteristics of NH₃/H₂/Air Flames

3.4.1 Sensitivity Analysis for Laminar Combustion Velocity

Figure 3-11 shows the sensitivity analysis with respect to laminar combustion velocity, indicating reactions where the present mech and Okafor Reduced Mech

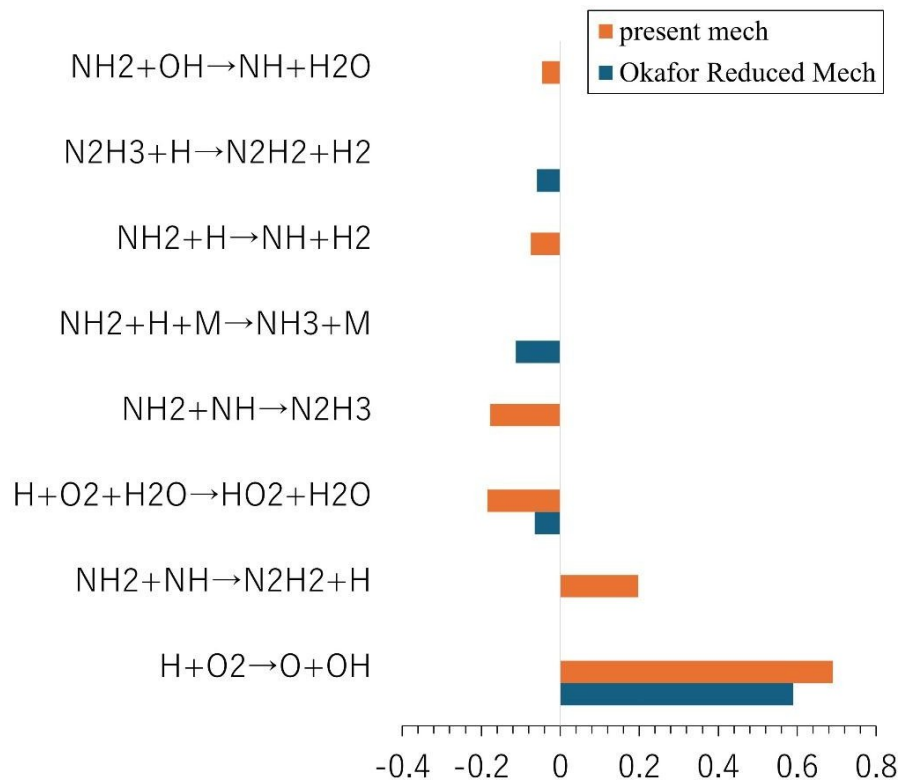


Figure 3-11 Sensitivity coefficients for laminar flame speeds of NH₃/H₂/Air flames predicted by present and Okafor reduced mechanisms

First, it was found that the reaction with the highest positive sensitivity in both mechanisms is $\text{H}+\text{O}_2\rightarrow\text{O}+\text{OH}$, indicating that the fundamental reaction governing laminar flame speed is common to both mechanisms.

In the Okafor Reduced Mechanism, the primary inhibitory reactions were $\text{NH}_2 + \text{H}^* + \text{M} \rightarrow \text{NH}_3 + \text{M}$ and $\text{N}_2\text{H}_3 + \text{H} \rightarrow \text{N}_2\text{H}_2 + \text{H}_2$. In contrast, in the present mechanism, these reactions disappear, and instead, $\text{NH}_2 + \text{NH} \rightarrow \text{N}_2\text{H}_3$ and $\text{NH}_2 + \text{NH} \rightarrow \text{N}_2\text{H}_2 + \text{H}$ are observed as strong inhibitory and promotional reactions, respectively. Furthermore, $\text{NH}_2+\text{NH}\rightarrow\text{N}_2\text{H}_2+\text{H}$, $\text{NH}_2+\text{NH}\rightarrow\text{N}_2\text{H}_3$, $\text{NH}_2+\text{H}\rightarrow\text{NH}+\text{H}_2$, and $\text{NH}_2+\text{OH}\rightarrow\text{NH}+\text{H}_2\text{O}$ (at the top of the graph) exhibited high sensitivity only in the present mechanism. These findings suggest that the updated NH_2 and N_2H_x -related submechanisms play a dominant role in the combustion characteristics of $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

Furthermore, it was found that the negative sensitivity of the three-body reaction $\text{H}+\text{O}_2+\text{H}_2\text{O}\rightarrow\text{HO}_2+\text{H}_2\text{O}$ has increased significantly.

. It is believed that this strong inhibitory reaction maintains equilibrium with NH_2 -related promotional reactions, contributing to the accurate prediction of laminar combustion rates.

Based on the above, it is believed that the **present mechanism** achieves high predictive accuracy relative to experimental values through the equilibrium between NH_2 and NH -related reactions and the three-body reaction that suppresses the reaction field.

3.4.2 Reaction Pathway Analysis

To identify the dominant factors in improving the prediction accuracy of laminar combustion rates, we conducted an analysis based on reaction pathway analysis and sensitivity analysis. Figure 3-12 shows the reaction pathway diagram obtained from **the present mechanism** under stoichiometric conditions for an $\text{NH}_3/\text{H}_2/\text{Air}$ flame ($\alpha/\text{H}_2=0.50$). A comparison of the reaction pathways between **the Okafor Reduced Mechanism** and **the present mechanism** revealed differences in the following three pathways.

First, the contribution of the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_3$ pathway is notable. **R3-8**, which was not considered in **the Okafor Reduced Mech**, appears as a significant pathway.



The sensitivity analysis results in Figure 3-11 also show that **R3-8** has a high negative sensitivity coefficient, indicating that the consumption of NH_2 and NH contributes to the suppression of the combustion rate. This suggests that the introduction of reactions such as **R3-8**, which contribute to the concentration of NH_x radicals, has improved the accuracy of predicting the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

Second, the activation of the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ pathway and the $\text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ pathway. As shown in Figure 3-12, the *IROP* of the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2$ pathway has increased significantly to 292.3% compared to the **Okafor Reduced Mechanism**. Furthermore, the *IROP* of the $\text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N}_2\text{H}_2$ pathway has increased substantially to 184.7% and 246.1%, respectively, compared to **the Okafor Reduced Mechanism**. Similarly, the *IROP* for the subsequent $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ pathway has increased significantly to 277.8% and 205.0%, respectively, compared to **the Okafor Reduced Mechanism**. In particular, reaction **R3-9**, the decomposition of NNH , directly generates H radicals, and it is believed that the increase in *IROP* for this pathway promotes the overall reaction in the flame.



Third, the importance of the pathway that generates N_2H_2 from NH_2 and NH was suggested.



R3-10 also exhibits a high positive sensitivity coefficient in the sensitivity analysis. Given that the *IROP* for the $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2$ pathway and

the $\text{NH} \rightarrow \text{N}_2\text{H}_2$ pathway, the *IROP* has increased significantly; this suggests that the enhancement of the H radical generation process via the previously

underestimated $\text{NH}_x \rightarrow \text{N}_2\text{H}_2$ pathway contributes to improving the prediction accuracy of the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

As described above, it is believed that reactions involving NH_2 , NH , N_2H_2 , N_2H_3 , and NNH are dominant in the laminar combustion rate of the $\text{NH}_3/\text{H}_2/\text{Air}$ flame. Next, to investigate the combustion characteristics of the $\text{NH}_3/\text{H}_2/\text{Air}$ flame, we examined the changes in the mole fractions of H , OH , NH_3 , NH_2 , NH , NNH , N_2H_2 , and N_2H_3 within the computational domain.

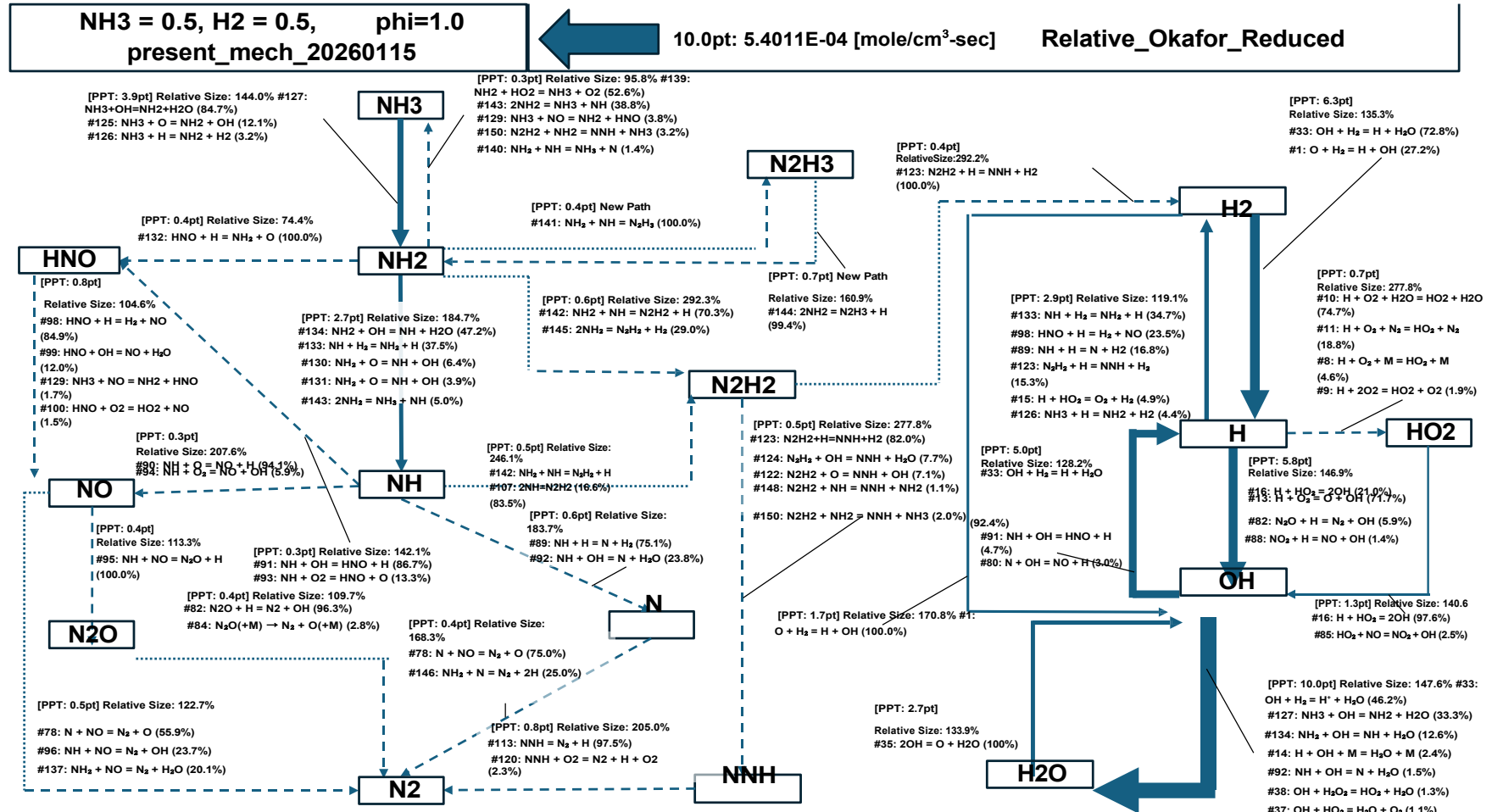


Figure 3-12 Reaction pathway of NH₃/H₂/air flames at
 $x_{H_2} \phi = 0.50$ and $\phi = 1.0$ calculated using the present mechanism

3.4.3 Comparison of mole fractions of chemical species

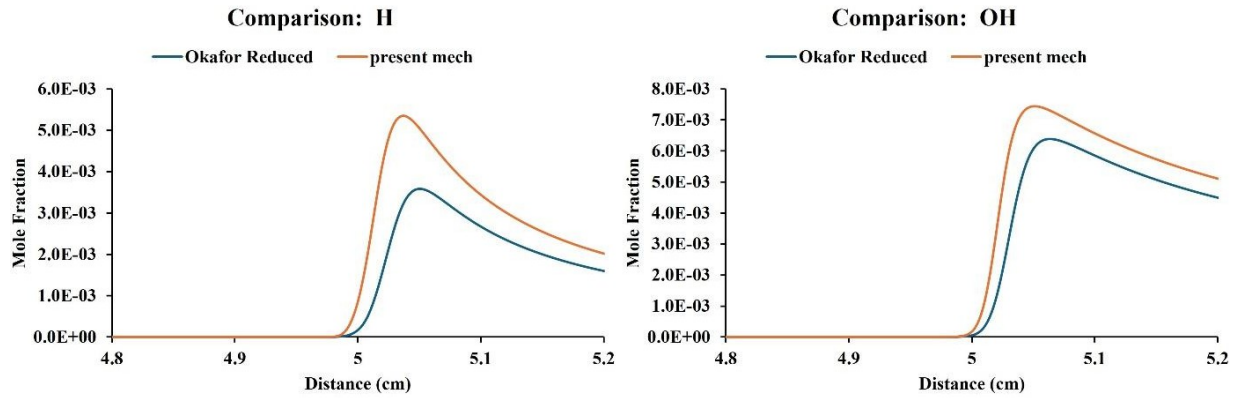


Figure 3-13 Mole fraction profiles of H and OH radicals in NH₃/H₂/Air flames at $\phi=1.0$ and $x_{H_2}=0.50$

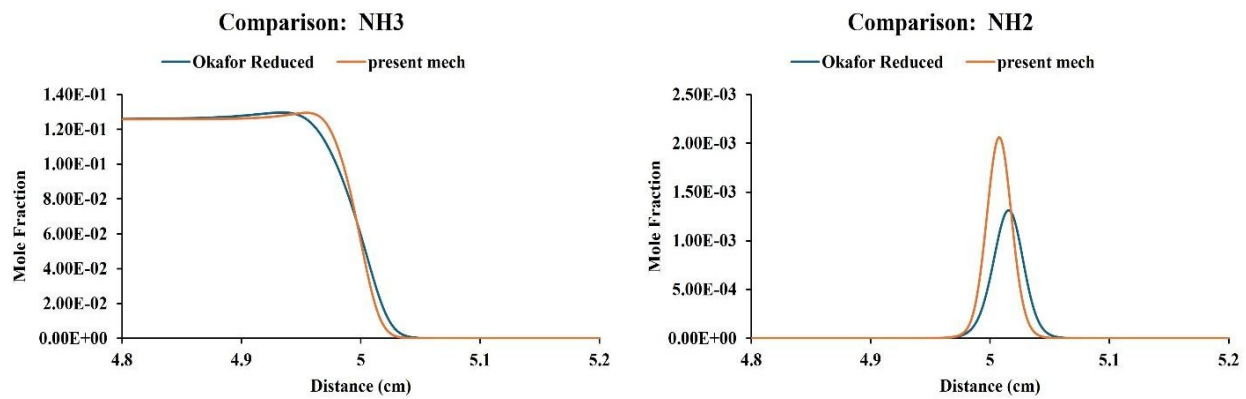


Figure 3-14 Mole fraction profiles of NH₃ and NH₂ radicals in NH₃/H₂/Air flames at $\phi=1.0$ and $x_{H_2}=0.50$

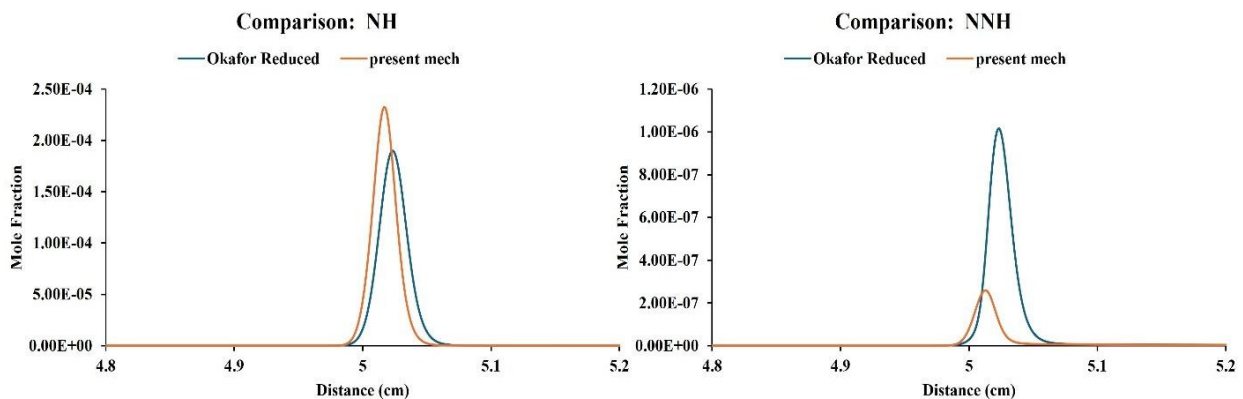


Figure 3-15 Mole fraction profiles of NH and NNH radicals in NH₃/H₂/Air flames at $\phi=1.0$ and $x_{H_2}=0.50$

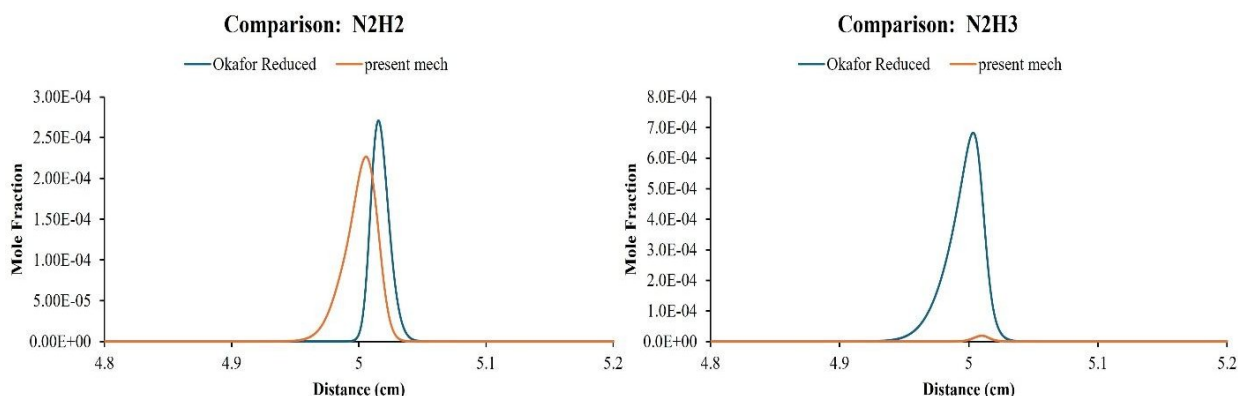


Figure 3-16 Mole fraction profiles of N_2H_2 and N_2H_3 radicals in $\text{NH}_3/\text{H}_2/\text{Air}$ flames at $\phi=1.0$ and $x/H_2=0.50$

Figures 3-13 to 3-16 show the changes in the mole fractions of H , OH , NH_3 , NH_2 , NH , NNH , N_2H_2 , and N_2H_3 in the computational domain of $\text{NH}_3/\text{H}_2/\text{Air}$ flames ($\phi=0.50$) under stoichiometric conditions. Overall, it can be seen that, compared to the Okafor Reduced model, the flame front is shifted slightly upstream in the present model.

As shown in Figure 3-13, the peak concentrations of the major radicals, H and OH , have increased significantly. This is thought to be due not only to the promotion of the $\text{H}+\text{O}_2\rightarrow\text{O}+\text{OH}$ chain-branching reaction indicated by the sensitivity analysis, but also to the enhancement of the H radical generation pathway via NH_x and N_2H_x intermediates. Consequently, as shown in Figure 3-14, the consumption rate of the fuel NH_3 increases, and the peak concentration of the NH_2 radical rises.

Next, focusing on NH_x and N_2H_x , characteristic differences between the two reaction mechanisms were identified. As shown in Figures 3-15 and 3-16, while the peak concentrations of NH and NH_2 increase, those of NNH , N_2H_2 , and N_2H_3 decrease compared to the Okafor Reduced system. In particular, a significant decrease is observed for NNH and N_2H_3 .

This finding is consistent with the results of the reaction pathway analysis. In the present mechanism, the I_{ROP} values for the $\text{NH}_2\rightarrow\text{N}_2\text{H}_2\rightarrow\text{NNH}\rightarrow\text{N}_2$ and $\text{NH}_2\rightarrow\text{NH}\rightarrow\text{N}_2\text{H}_2\rightarrow\text{NNH}\rightarrow\text{N}_2$ pathways have increased by approximately 200–300% compared to the Okafor Reduced Mechanism. In other words, this implies that while nitrogen intermediates such as NNH and N_2H_x are generated in the present mechanism, their consumption reactions proceed even more rapidly. In particular, since the rapid consumption of NNH and N_2H_x directly leads to the release of H radicals that promote the overall flame reaction, it can be concluded that this contributes to an increase in laminar combustion rate even as the concentration of nitrogen intermediates is suppressed. As described above, from the perspective of flame structure, it has been confirmed that the present mechanism improves upon the underestimation of the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames—a shortcoming of the Okafor Reduced Mechanism—through the optimization of nitrogen intermediate-related submechanisms. In the next section, we will perform a sensitivity analysis on chemical species to identify the reactions that are particularly dominant in the formation and consumption of NH_x and N_2H_x in the present mechanism and the Okafor Reduced Mechanism.

was performed.

3.4.4 Sensitivity analysis with respect to the mole fraction of chemical species

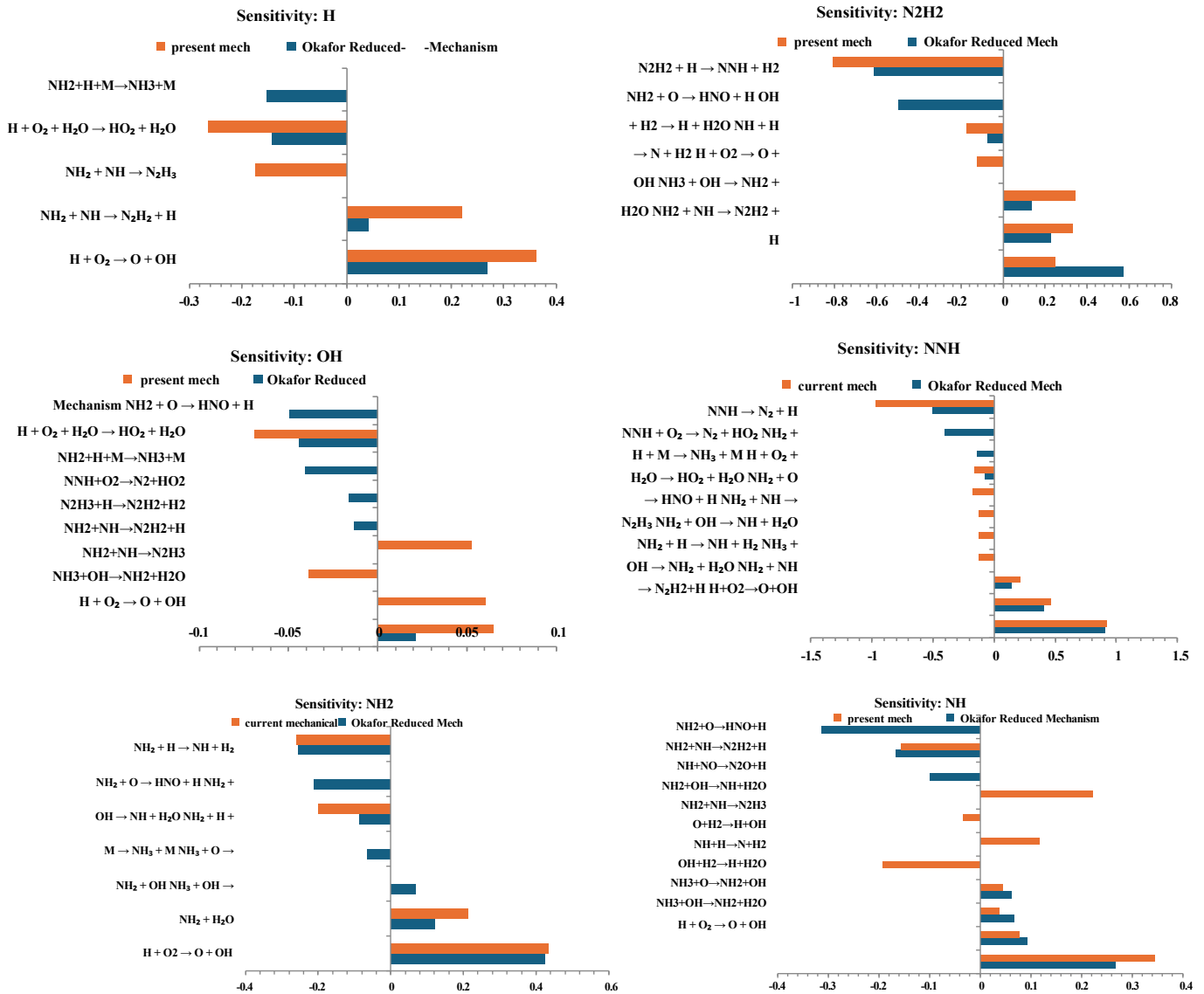


Figure 3-17 Sensitivity coefficients for the mole fractions of H, OH, NH₂, NH, NNH, and N₂H₂ for NH₃/H₂/air flames at $\phi = 1.0$ and $x_{\text{H}_2} = 0.50$

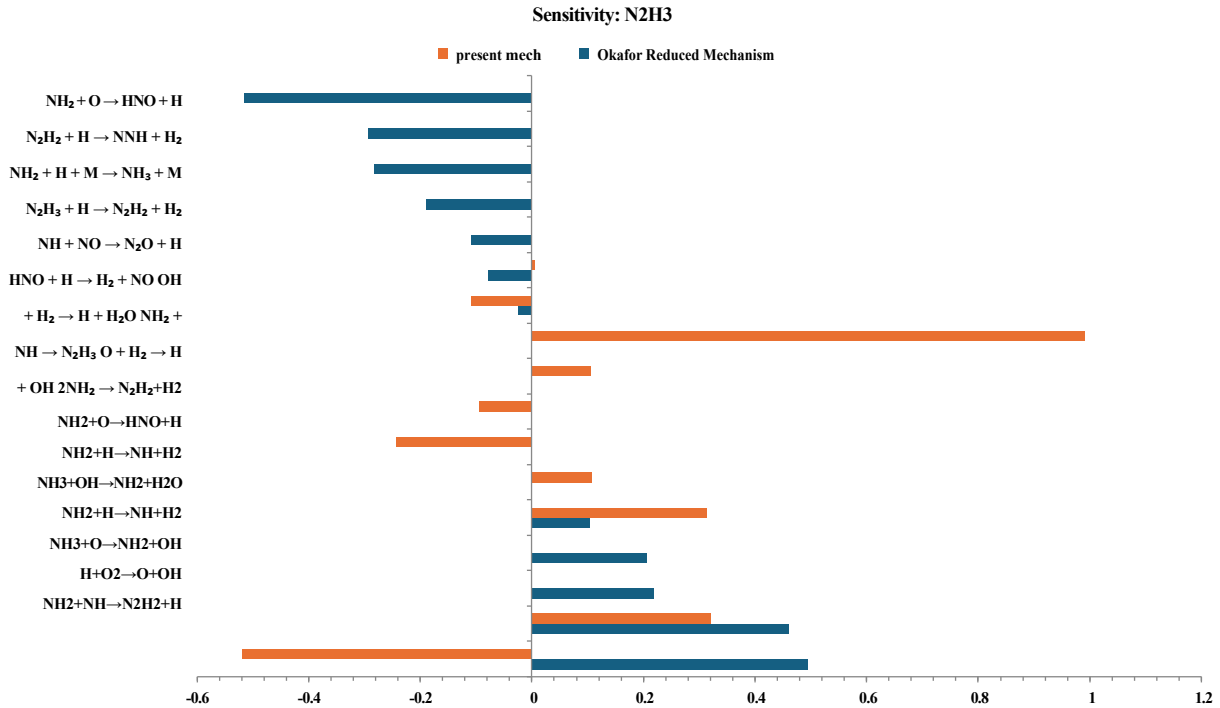


Figure 3-18 Sensitivity coefficients for N_2H_3 mole fractions in $\text{NH}_3/\text{H}_2/\text{Air}$ flames at $\phi=1.0$ and $x/\text{H}_2=0.50$

To identify the reactions governing the behavior of intermediate species in $\text{NH}_3/\text{H}_2/\text{Air}$ flames ($x/\text{H}_2=0.50$) under stoichiometric conditions, a sensitivity analysis was performed for the major chemical species. Figures 3-17 and 3-18 show the reactions with particularly significant differences in the sensitivity analysis results for the mole fractions of H, OH, NH_2 , NH, NNH, N_2H_2 , and N_2H_3 in $\text{NH}_3/\text{H}_2/\text{Air}$ flames ($\phi=0.50$) under stoichiometric conditions.

First, regarding the sensitivity analysis of H and OH radicals, the chain-branching reaction $\text{H} + \text{O}_2 \rightarrow \text{O} + \text{OH}$ showed the greatest positive sensitivity in both reaction mechanisms and promoted the overall reaction. Furthermore, this reaction exhibits a large positive sensitivity regarding the formation of other chemical species. On the other hand, the three-body reaction ' $\text{H} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{HO}_2 + \text{H}_2\text{O}$ ' dominates the negative sensitivity. In particular, in the '**present mech**', the negative sensitivity of this reaction is significantly increased compared to the '**Okafor Reduced Mech**', strongly indicating chain termination by H_2O .

Next, we focus on the behavior of NH_2 and NH radicals. First, the formation of NH_2 is primarily governed by the reaction $\text{NH}_3 + \text{OH} \rightarrow \text{NH}_2 + \text{H}_2\text{O}$, which occurs mainly in the early stages of fuel consumption. Regarding the consumption reactions of NH_2 , $\text{NH}_2 + \text{H} \rightarrow \text{NH} + \text{H}_2$ is the major pathway in both mechanisms; however, in the "**present mech**" mechanism, the contribution of $\text{NH}_2 + \text{OH} \rightarrow \text{NH} + \text{H}_2\text{O}$ is particularly significant. Looking at the sensitivity analysis results for both NH_2 and NH radicals, we see that the $\text{NH}_2 + \text{OH}$ reaction exhibits a large negative sensitivity (consumption) for NH_2 , whereas it exhibits a large positive sensitivity (generation) for NH. In other words, in the **present mechanism**, this reaction consumes NH_2 radicals while strongly promoting the generation of NH radicals. Furthermore, focusing on the behavior of the generated NH radicals, in the **present mechanism**, $\text{NH} + \text{H} \rightarrow$

N+H₂ exhibits a large negative sensitivity toward NH and functions as the primary decomposition pathway that consumes NH and converts it to N.

Next, we focus on the N₂H_x species. It is evident that these intermediate species depend heavily on the recombination of NH_x radicals. For N₂H₂ and NNH, the reaction NH₂+NH→N₂H₂+H exhibits a dominant positive sensitivity, and the sensitivity analysis results for the H radical also indicate that this is the primary reaction leading to the formation of the H radical. On the other hand, for N₂H₃, the reaction NH₂+NH→N₂H₂+H exhibits the highest negative sensitivity, indicating that it is the dominant reaction for the consumption of N₂H₃. Furthermore, for N₂H₃, it is evident that NH₂+NH→N₂H₃ exhibits overwhelmingly positive sensitivity. It is worth noting that NH₂+NH→N₂H₃ is not included in the Okafor Reduced Mech and was added during optimization.

Based on the above analysis, the main improvements in the present mechanism over the Okafor Reduced Mechanism can be summarized as “enhancement of the inhibitory effect of H₂O” and “optimization of the NH_x recombination reaction.”

First, although a large amount of H₂O is present as a product under H₂ co-combustion conditions, the negative sensitivity of the reaction H₂O + H₂O → H₂O + H₂O in the present mechanism has increased significantly compared to the Okafor Reduced Mechanism, indicating that it functions as a powerful suppressing reaction.

Second, the NH₂ + NH reaction is described in detail in the present mechanism. Sensitivity analysis revealed that the NH₂ + NH recombination reaction plays two major roles. The first is to activate the reaction site through H radical generation via R3-11



Second, it suppresses the activation of the reaction site by generating N₂H₃ via R3-12 and scavenging NH₂ and NH radicals.



In the present model, we have added an N₂H₃ formation pathway—which significantly contributes to the suppression of the entire reaction field but was not considered in the Okafor Reduced- Mechanism—while simultaneously increasing the sensitivity of the N₂H₂ pathway, which contributes to the activation of the entire reaction field. These changes are believed to have optimized the balance between promotion and suppression, thereby improving the predictive accuracy of laminar combustion rates.

3.4.5 Important submechanisms in NH₃/H₂/Air flames

As mentioned in the previous section, it was suggested that in **the present model**, the newly added reaction R3-13 and R3-14, for which the reaction rate constant was updated, contribute to the accurate prediction of the laminar combustion rate.



However, based on the results of the sensitivity analysis alone, it was not possible to identify quantitative changes in whether the reaction rate itself was physically accelerated or inhibited by the refinement of the reaction mechanism. Therefore, for reactions considered to have a particularly significant impact on the NH₃/H₂/Air, we investigated the changes in reaction rate constants before and after the modification. Figure 3-19 shows a comparison of the reaction rate constants (k) for R3-13 between **the present mechanism** and **the Okafor Reduced Mechanism**. Note that T in the figure represents the flame temperature [K].

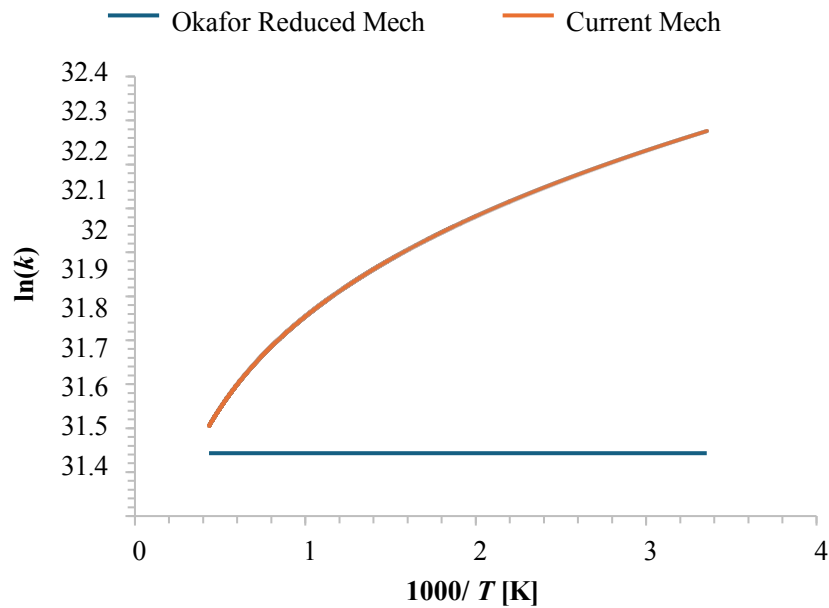


Figure 3-19 Comparison of rate constants for the reaction $\text{NH}_2 + \text{NH} = \text{N}_2 \rightarrow \text{H}_2 + \text{H}$ between the Okafor Reduced Mechanism and the present mechanism

As shown in Figure 3-19, the **Okafor Reduced Mechanism** exhibits low temperature dependence, and the values are set lower than those of **the present mechanism** across the entire temperature range. In contrast, the rate constants based on reaction rate constants based on S.J. Klippenstein, et al. (2009) [27] show higher values than those in the Okafor Reduced Mech, which uses data from the Tian Mech [25]. The difference is particularly pronounced in the low-temperature range.

The increase in the rate constant of **R3-13** is consistent with the results of the sensitivity analysis, which showed that this reaction exhibited a greater positive sensitivity toward **H** radical formation. However, simply increasing the rate constant may lead to an overestimation of the laminar combustion rate. Therefore, the suppressing effect of **R3-14** is crucial. While **R3-14** was not considered in the **Okafor Reduced Mechanism**, in the **present mechanism**, it is thought to function as a pathway for **N₂H₃** formation while also playing a role in suppressing excessive activation of the reaction field by competing with **R3-13**.

Furthermore, to analyze the characteristics of the **present mechanism**, we focused on the behavior of **N₂H₂** and **NNH**, which are generated following the **NH₂+NH** reaction. Based on the results of the sensitivity analysis, we identified the following reactions, **R3-15** and **R3-16**, as having a dominant influence on the mole fractions of these species.



The sensitivity analysis results suggest that **R3-15** and **R3-16** have the highest negative sensitivity to the mole fractions of **N₂H₂** and **NNH**, respectively, and are the reactions that govern their consumption. Figures 3-20 and 3-21 show a comparison of the reaction rate constants for the **present mechanism** and the **Okafor reduced mechanism** in these two reactions.

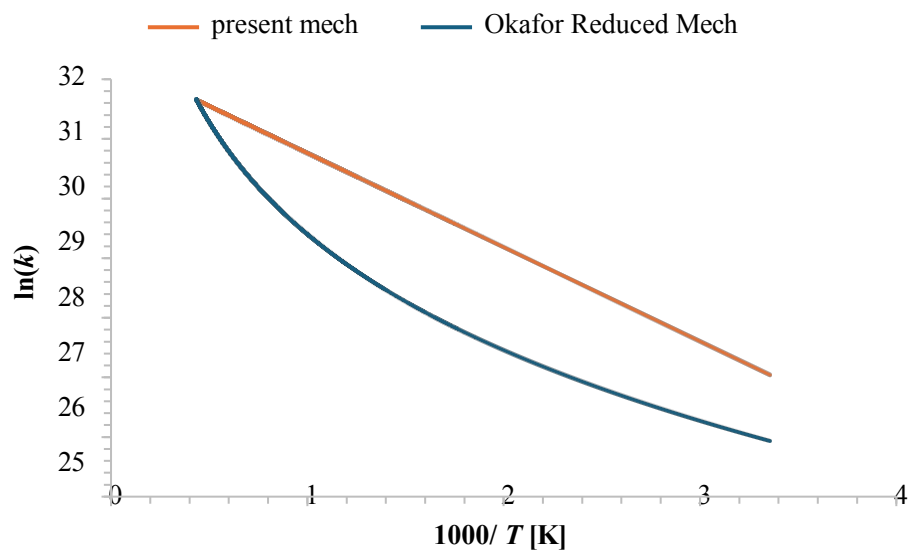


Figure 3-20 Comparison of rate constants for the reaction $\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$ between the Okafor-reduced mechanism and the present mechanism

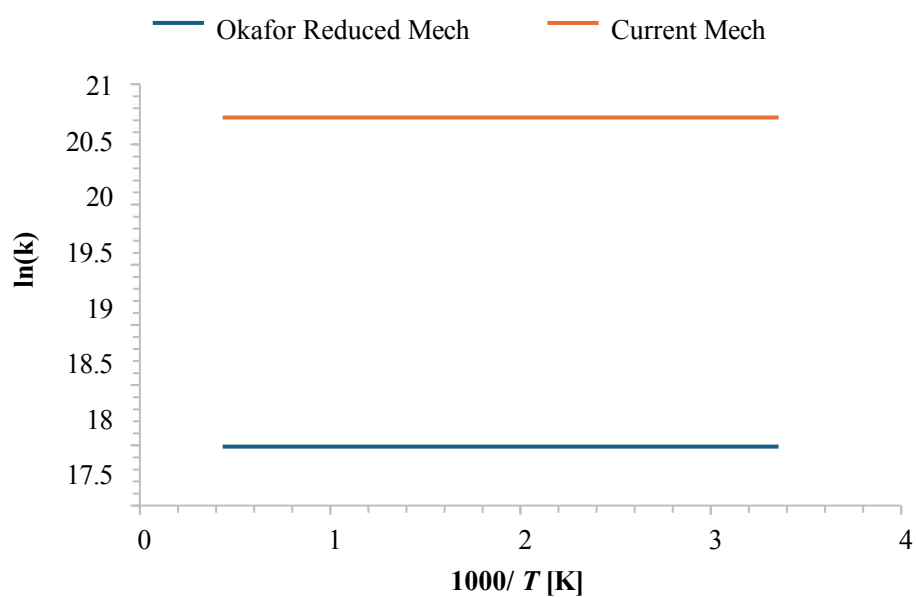


Figure 3-21 Comparison of rate constants for the reaction $\text{NNH} = \text{N}_2 + \text{H}$ between the Okafor Reduced Mechanism and the present mechanism

First, we focus on R3-15, the primary consumption reaction of N_2H_2 . The **present mechanism** adopts the rate constants based on P. Glarborg, et al. (2018)[28]. Compared to the **Okafor Reduced Mech**, which uses data from the **Tian Mech**, it can be seen in Figure 3-20 that the present mech has higher reaction rate constants across the entire temperature range. In particular, the difference widens in the low- to mid-temperature range.

Furthermore, for R3-16, which is an NNH consumption reaction and directly supplies H radicals to the reaction site, the difference in rate constants is even more pronounced. As can be seen from Figure 3-21, the reaction rate constant in the **present Mech model**, which uses data from S.J. Klippenstein et al. (2011)[27], the reaction rate constant in the **present mech** is set approximately 3 points higher in $\ln(k)$ compared to the **Okafor Reduced Mech**, which uses data from **Tian Mech**.

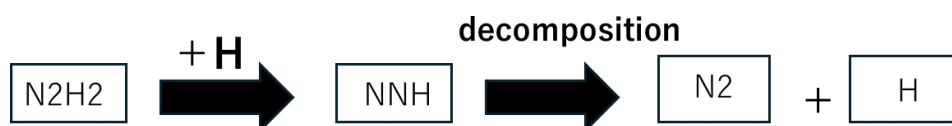


Figure 3-22 Conversion processes of N_2H_x species

The update of the reaction rate constants described above revealed that, in the **present mech**, the series of conversion processes $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2 + \text{H}$ shown in Figure 3-22 has been enhanced. These reactions rapidly convert N_2H_x intermediates to N_2 while simultaneously playing a role in directly generating and supplying H radicals that activate the overall combustion reaction, and this is thought to contribute significantly to the combustion characteristics of the **present mechanism**.

4. Conclusion Conclusion

In this study, we proposed a reaction mechanism, “**present mech.**” capable of accurately predicting laminar combustion rates simultaneously under both $\text{NH}_3/\text{H}_2/\text{Air}$ and $\text{NH}_3/\text{CH}_4/\text{Air}$ flame conditions, which had not been reported previously. The calculated laminar combustion rates for $\text{NH}_3/\text{H}_2/\text{Air}$ and $\text{NH}_3/\text{CH}_4/\text{Air}$ flames using the **reaction mechanism** proposed in this study agree well with literature values, confirming the validity of the reaction mechanism. Furthermore, reaction pathway analysis identified the reactions that contribute significantly to the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames. Specifically, it was found that the competition between the promotion of H radical generation during NH_x recombination and NNH consumption reactions and the suppressing effect on the entire reaction field due to N_2H_3 formation contributes to improving the prediction accuracy of the laminar combustion rate of $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

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Although I caused some trouble due to my shortcomings, I am truly grateful for all your support.

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Since this was my primary research topic, I owe a special debt of gratitude to Professor Okafo. I can't thank him enough for staying up past midnight to guide me as I wrote my paper for the Internal Combustion Symposium. Beyond my research, I also have fond memories of bumping into him and his son at Tenjin Central Park while I was working my part-time job at Red Bull.

Also, although I didn't have much direct interaction with him due to the nature of my work in numerical computation, I am truly grateful to Mr. Fujisaki, a technical staff member who always shared such entertaining stories at our drinking gatherings. The time I spent drinking with Mr. Fujisaki at the fishermen's hut remains a fond memory.

Although it was only a short year, Professor Matsuda was always approachable—whether we were going to the cafeteria together or when he enthusiastically guided my research—and I am truly grateful for everything you did for me.

Finally, I would like to take this opportunity to express my gratitude to everyone who was involved with me during my six years as a student, as well as to my parents and younger brother, who always supported me.

February 4, 2026,
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Energy Systems
Shunnosuke Abe

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